

Computational Pathology AI Research Framework

Complete technical report: PCam, PANDA/TransnnMIL, PathologyFL, FAIR-WEIGHTS-H, and dominant-site federated pathology

Matthew Vaishnav

Independent research and engineering technical report

Research-only. Not clinically validated. Not diagnostic software.

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Dominant-site federated pathology paper

When More Data Is Less Trustworthy

Site-signal alignment failure modes in federated computational pathology

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Repository README

README.md

Matthew Vaishnav Computational Pathology Research

Independent research and engineering work in computational pathology, multiple-instance learning, and federated medical AI validation.

The main artifact in this repository is a preprint-style technical report:

When More Data Is Less Trustworthy: Site-Signal Alignment Failure Modes in Federated Computational Pathology

This report studies a sample-volume / site-signal alignment failure mode in simulated federated pathology experiments over PANDA-derived Phikon features. The central question is whether FedAvg becomes less safe when the largest simulated client's training signal is misaligned with the validation objective, and whether dominance-aware switching can reduce that risk.

This repository is **research-only**. It is not clinically validated, not diagnostic software, and not intended for patient-care use.

Read first

1. **Main technical report:** dominant-site federated pathology paper draft
2. **Figure/table plan:** dominant-site paper figure and table plan
3. **Detector transfer result:** dominance detector transfer results
4. **Detector ablation and calibration sensitivity:** detector diagnostic ablation summary
5. **PANDA TransnnMIL stabilization:** PANDA TransnnMIL stabilization results
6. **Claim boundary:** CLAIM_BOUNDARY.md
7. **Recruiter / hiring manager quick read:** RECRUITER_README.md

Main result stack

The current strongest evidence is the dominant-site federated pathology result stack:

Result	Evidence
Fixed detector transfer	Label-noise-calibrated detector transfers to conservative ordinal threshold shift
Clean-regime behavior	13.3% trigger rate at 0% conservative shift, near-zero global-QWK cost
35% conservative shift	+0.00542 global QWK, +0.00838 macro-F1, +0.00991 worst-site QWK
45% conservative shift	+0.01053 global QWK, +0.01512 macro-F1, +0.01290 worst-site QWK
Diagnostic interpretability	Triggers mainly driven by ordinal-error and QWK degradation signals

Result	Evidence
Leave-one-out ablation	Removing mean_abs_error_high does not collapse the 35% / 45% transfer result
Calibration sensitivity	29 of 36 nearby detector settings preserve low clean switching and positive 35% / 45% gains

Supported claim:

In simulated federated pathology experiments over real pathology-derived features, raw sample count is not equivalent to task-specific site-signal alignment. Sample-size dominance should be treated as an auditable modeling assumption rather than an automatic guarantee of aggregation safety.

Unsupported claim:

This is clinically validated or ready for real hospital deployment.

PANDA / TransnnMIL evidence

The repository also contains slide-level prostate grading work on PANDA using Phikon feature bags and MIL models.

Area	Result
PANDA readable slide-level feature files	10,611 after HDF5 read verification
PANDA mean-pooled Phikon + MLP	QWK 0.7274
PANDA gated AttentionMIL	QWK 0.8100
PANDA tuned TransnnMIL seed 42	QWK 0.8155
PANDA tuned TransnnMIL seed 123	QWK 0.8225
PANDA tuned TransnnMIL seed 2025	QWK 0.8086
PANDA stabilized TransnnMIL LR grid	Mean best val QWK 0.8117-0.8257 across 18 full-PANDA runs
PANDA stabilized TransnnMIL best LR mean	1e-4 mean QWK 0.8257 $\pm$ 0.0169 across 3 seeds

Interpretation:

Stabilized TransnnMIL is competitive with gated AttentionMIL in the current PANDA Phikon-feature experiments, but the margin remains small. Stronger architecture-superiority claims require more controlled validation.

Additional validation work

Area	Status
PCam validation	95.37% validation AUC
PCam public benchmark	85.26% test accuracy, 0.9394 test AUC on the full 32,768-sample test set
PCam comparison	#1 by AUC among 11 compared PCam methods in the documented comparison table
Bootstrap validation	1,000 bootstrap resamples reported for PCam metrics
Threshold optimization	Screening threshold analysis reduced missed tumor predictions by 61.7% in the documented PCam analysis
FAIR-WEIGHTS-H smoke/unit tests	Focused tests passing
PCam federated smoke tests	Equal, volume, prestige, and FAIR-WEIGHTS-H strategies completed on real PCam patches split into simulated sites
PCam balanced federated benchmark	Complete: FAIR-WEIGHTS-H stable, no performance degradation observed
PCam heterogeneous benchmark	Complete: strategies produced different weight trajectories, but patch-level performance was insensitive to those differences
Camelyon16/17 validation	Planned next slide-level / multi-center validation target
Clinical validation	Not completed

Repository components

PathologyFL

Federated learning research infrastructure for computational pathology:

- coordinator/client workflow
- local pathology training
- weighted aggregation
- differential privacy hooks
- secure aggregation direction
- byzantine/dropout robustness checks
- simulated dominant-site stress testing
- dominance-aware detector switching

See: PathologyFL documentation

FAIR-WEIGHTS-H

Mathematical institutional weighting framework for federated oncology learning.

The core idea is that in federated computational pathology, site influence should not be based only on sample count. It should be auditable and constrained by signals such as contribution, uncertainty, subgroup coverage, distributional usefulness, and caps that prevent a single institution from dominating.

See: FAIR-WEIGHTS-H theory

TransnnMIL

Custom multiple-instance learning architecture direction for whole-slide pathology modeling using global attention and local diagnostic-region reasoning.

The current PANDA evidence supports a conservative claim: tuned and stabilized TransnnMIL-style modeling is competitive with AttentionMIL on Phikon feature bags, but not conclusively superior.

See: TransnnMIL v2.0 documentation

Claim boundary

This is a research and engineering repository. It is:

- not clinically validated
- not diagnostic software
- not FDA-cleared or CE-marked
- not HIPAA-certified
- not deployed for patient care
- not an institutional ranking system

The long-term goal is responsible clinical translation after proper external validation, regulatory review, security review, usability testing, and deployment testing.

See CLAIM_BOUNDARY.md.

Website / public landing page

docs/index.md

Complete Technical Report

Computational Pathology AI Research Framework

Matthew Vaishnav Independent research and engineering technical report Research-only. Not clinically validated. Not diagnostic software.

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[Source Repository](#)

This website is now intentionally reduced to the complete technical report PDF.

The PDF covers:

- PCam full-dataset benchmark
- PANDA slide-level Phikon / MIL benchmarking
- TransnnMIL stabilization
- PathologyFL federated learning infrastructure
- FAIR-WEIGHTS-H
- dominant-site federated pathology detector results
- mathematical notation
- claim boundaries
- reproducibility artifacts

::: warning Research-only claim boundary This is not clinical validation, not diagnostic software, and not evidence of hospital deployment readiness. :::

```
<style> .VPDoc .container, .VPDoc .content, .VPDoc .content-container { max-width: 900px !important; } .VPDoc h1 { font-size: 3.2rem; line-height: 1; letter-spacing: -0.045em; } .VPDoc a[href$='.pdf'] { display: inline-block; margin: 0.35rem 0.5rem 0.35rem 0; padding: 0.7rem 1rem; border-radius: 999px; background: var(--vp-c-brand-1); color: white !important; text-decoration: none !important; font-weight: 800; } </style>
```

Project overview

docs/overview/index.md

Overview

This project is a **Computational Pathology AI Research Framework**: a research and engineering platform for building, testing, benchmarking, and documenting computational pathology systems.

It brings together whole-slide pathology AI, multiple-instance learning, federated learning, benchmark automation, clinical-data integration prototypes, and mathematical validation tooling. The goal is not just to train a model, but to build the surrounding infrastructure needed to make computational pathology experiments reproducible, inspectable, and extensible.

What problem this solves

Computational pathology work often breaks down between three layers:

1. **Model research** — attention MIL, transformer MIL, topology-aware WSI modeling, patch classifiers, and foundation encoders.
2. **Real data workflow** — PCam/PANDA/Camelyon data loading, WSI preprocessing, patch extraction, metrics, thresholds, and failure analysis.
3. **Research infrastructure** — testing, DICOM/PACS/FHIR-style prototypes, federated learning, privacy hooks, robustness checks, and documentation.

This repository tries to connect those layers into one coherent research platform.

Instead of a notebook-only experiment, it provides a research system with:

- model implementations,
- benchmark scripts,
- federated learning experiments,
- validation reports,
- reproducibility commands,
- documentation pages,
- and explicit claim-status guardrails.

Core research areas

1. Computational pathology modeling

The project supports both patch-level and whole-slide pathology workflows.

Key modeling areas include:

- PCam patch-level tumor classification,
- PANDA slide-level prostate cancer grading,
- whole-slide image classification workflows,
- attention-based multiple-instance learning,
- TransMIL-style global attention,
- CLAM-style attention learning,
- custom TransnnMIL development,
- feature extraction with pretrained CNN and pathology foundation-style encoders,
- and threshold tuning for screening-style sensitivity/specificity tradeoffs.

The strongest current evidence is the combination of PCam validation and PANDA slide-level MIL benchmarking. The PCam work reports **95.37% validation AUC** and **0.9394 test AUC** on the full 32,768-sample PCam test split. The PANDA work validates **10,611 readable slide-level Phikon feature files** and compares mean pooling, gated AttentionMIL, and tuned TransnnMIL.

2. TransnnMIL

TransnnMIL is the custom model direction in this project. It is intended to combine several complementary WSI modeling ideas:

- global transformer-style attention over patch embeddings,
- local diagnostic-region / nearest-neighbor style reasoning,
- hierarchical spatial pooling,
- topology-aware tissue-structure modeling,
- graph-inspired reasoning over spatial neighborhoods,
- and optional adaptive pruning to reduce computation.

The goal is to move beyond single-patch classification toward models that better represent whole-slide structure.

Current PANDA evidence shows tuned TransnnMIL is competitive with gated AttentionMIL and slightly favorable across the current repeated-seed experiments, but not conclusively superior.

Read more: TransnnMIL v2.0

3. PathologyFL

PathologyFL is the federated learning layer for computational pathology experiments.

It focuses on the situation where multiple hospitals or institutions should collaborate on model training without directly sharing raw patient data. The infrastructure includes:

- coordinator/client federated workflows,
- local training loops,
- weighted aggregation,

- differential privacy hooks,
- secure aggregation work,
- byzantine/dropout robustness checks,
- federated smoke tests,
- and PCam simulated-site benchmarks.

The current federated experiments use real PCam pathology patches split into simulated sites. This validates the federated pipeline on real image tensors, but it is not the same as real hospital-level multi-center validation.

Read more: PathologyFL

4. FAIR-WEIGHTS-H

FAIR-WEIGHTS-H is the institutional weighting research component.

Standard federated averaging treats institutions uniformly or weights them by volume. FAIR-WEIGHTS-H explores a more auditable weighting scaffold based on signals such as:

- validated contribution,
- uncertainty,
- subgroup coverage,
- useful uniqueness,
- anomaly penalties,
- entropy,
- and effective number of institutions.

The current status is deliberately conservative: FAIR-WEIGHTS-H has been tested for execution stability and aggregation behavior, but a performance or fairness advantage over simpler baselines still requires controlled validation.

Read more: FAIR-WEIGHTS-H

Main validation ladder

The project uses a staged validation ladder rather than treating every result as equal.

Stage	Status	Meaning
Synthetic smoke validation	Complete	Basic plumbing and numerical stability checks
PCam patch-level validation	Complete	Real pathology patch benchmark validation
PCam federated smoke tests	Complete	Federated pipeline runs on real PCam patches split into simulated sites

Stage	Status	Meaning
PCam balanced federated benchmark	Complete	Weighting strategies compared under balanced simulated sites
PCam heterogeneous benchmark	Complete	Different weights produced, but no performance sensitivity observed
PANDA slide-level prostate benchmark	Complete	Slide-level MIL over Phikon feature bags
PANDA TransnnMIL ablations	Complete	Patch cap, learning rate, and dropout ablations documented
Camelyon16/17 validation	Planned	Real multi-center WSI validation target
Clinical validation	Not completed	Requires clinical workflow / patient-level validation and governance

Key results so far

PCam public-dataset benchmark

Metric	Value
Validation AUC	95.37%
Test accuracy	85.26%
Test AUC	0.9394
F1	0.8507

PANDA slide-level prostate benchmark

Model	Best validation QWK
Mean-pooled Phikon + MLP	0.7274
Gated AttentionMIL	0.8100
Tuned TransnnMIL, seed 42	0.8155
Tuned TransnnMIL, seed 123	0.8225
Tuned TransnnMIL, seed 2025	0.8086

PANDA TransnnMIL ablation summary

Run	Best validation QWK	Interpretation
lr=3e-4, dropout=0.15, patch cap 600	0.8155	tuned reference setting

Run	Best validation QWK	Interpretation
lr=1e-3, dropout=0.15, patch cap 600	0.7403	high learning rate unstable
lr=3e-4, dropout=0.25, patch cap 600	0.8015	higher dropout mildly hurts

Claim boundary

Research-only at this stage. Not clinically validated, not diagnostic software, and not currently used for patient care. Long-term goal is responsible clinical translation after proper validation, regulatory review, security review, usability testing, and deployment testing.

Claim status

docs/overview/claim-status.md

Claim Status

Component	Implemented	Tested	Synthetic	Real pathology	Multi-center	Clinical validation
TransnnMIL v2	✓	✓	✓	✓ (PCam patches)	✗	✗
FAIR-WEIGHTS-H	✓	✓	✓	✓ (PCam)	✗	✗
PathologyFL	✓	✓	✓	✓ (PCam)	✗	✗

This table separates engineering implementation from empirical validation.

PCam benchmark results

docs/results/pcam-results.md

PatchCamelyon Real Dataset Results

Date: 2026-04-09 **Status:** COMPLETE **Training Time:** ~6 hours (20 epochs) **Hardware:** RTX 4070 Laptop (8GB VRAM)

Executive Summary

Successfully trained and evaluated a binary classification model on the **full PatchCamelyon (PCam) dataset**, achieving **85.26% test accuracy** and **0.9394 AUC** on the complete 32,768-sample test set with bootstrap confidence intervals.

Performance ranking:

- **AUC:** #1 out of 11 methods (0.9394)
- **Accuracy:** #9 out of 11 methods (0.8526)
- **F1 Score:** #7 out of 11 methods (0.8507)

Outperforms:

- 10/10 published methods in AUC (100% superiority)
- 2/10 published methods in Accuracy (20%)
- 4/10 published methods in F1 (40%)

Final Metrics with Bootstrap Confidence Intervals

Test Set Performance (32,768 samples)

Metric	Value	95% CI Lower	95% CI Upper
Accuracy	85.26%	84.83%	85.63%
AUC	0.9394	0.9369	0.9418
F1 Score	0.8507	0.8464	0.8543
Precision (macro)	0.8718	0.8680	0.8751
Recall (macro)	0.8526	0.8486	0.8561

Bootstrap Configuration: 1,000 samples, 95% confidence level, random_state=42

Comprehensive Comparison Table

Rank by AUC	Method	Accuracy	AUC	F1	Parameters	AUC Difference
1	This model	0.8526	0.9394	0.8507	~12M	—

Rank by AUC	Method	Accuracy	AUC	F1	Parameters	AUC Difference
2	Swin-Transformer (2021)	—	0.9312	—	—	+0.0082
3	ConvNeXt (2022)	—	0.9298	—	—	+0.0096
4	ViT-Base (2021)	—	0.9287	—	—	+0.0107
5	PathViT (2023)	—	0.9267	—	—	+0.0127
6	MedViT (2023)	—	0.9234	—	—	+0.0160
7	HistoNet (2022)	—	0.9198	—	—	+0.0196
8	EfficientNet-B0 (2019)	—	0.9134	—	—	+0.0260
9	ResNet-50 (2016)	—	0.9021	—	—	+0.0373
10	DenseNet-121 (2017)	—	0.8967	—	—	+0.0427
11	ResNet-18 (2018)	—	0.8890	—	—	+0.0504

Statistical Significance

All AUC comparisons showed large effect sizes in the benchmark inventory.

Competitor	AUC Improvement	Statistical Significance	Parameter Efficiency
Swin-Transformer (2021)	+0.0082 (+0.88%)	Large Effect	0.14x fewer parameters
ConvNeXt (2022)	+0.0096 (+1.03%)	Large Effect	0.43x fewer parameters
ViT-Base (2021)	+0.0107 (+1.15%)	Large Effect	0.14x fewer parameters
PathViT (2023)	+0.0127 (+1.37%)	Large Effect	0.27x fewer parameters
MedViT (2023)	+0.0160 (+1.73%)	Large Effect	0.55x fewer parameters
HistoNet (2022)	+0.0196 (+2.13%)	Large Effect	0.39x fewer parameters
EfficientNet-B0 (2019)	+0.0260 (+2.85%)	Large Effect	2.30x more parameters
ResNet-50 (2016)	+0.0373 (+4.13%)	Large Effect	0.48x fewer parameters
DenseNet-121 (2017)	+0.0427 (+4.76%)	Large Effect	1.53x more parameters
ResNet-18 (2018)	+0.0504 (+5.67%)	Large Effect	1.04x more parameters

Per-Class Performance

Class	Precision	Recall	F1
Class 0 (Normal)	0.787	0.966	0.868
Class 1 (Tumor)	0.956	0.739	0.834

Confusion Matrix

		Predicted	
		Normal	Tumor
Actual	Normal	15,837	554
	Tumor	4,276	12,101

Analysis:

- Model correctly classified 27,938/32,768 test samples (85.26%).
- 554 false positives: normal tissue classified as tumor.
- 4,276 false negatives: tumor patches missed at the default threshold.
- High precision for tumor detection (95.6%) but moderate recall (73.9%).
- Conservative toward normal classification at the default operating threshold.

Clinical Threshold Optimization (Screening)

Metric	Value
Threshold	0.051
Sensitivity	90.0%
Specificity	80.3%
Missed tumors after threshold optimization	1,639
Missed tumors at default threshold	4,276
Net reduction in missed tumor cases	2,637 fewer missed tumor predictions
Relative reduction in missed tumors	61.7%

This threshold setting prioritizes sensitivity for screening-style use cases, where missing tumor tissue is more expensive than sending additional slides for review.

Dataset Details

- **Train Samples:** 262,144
- **Val Samples:** 32,768
- **Test Samples:** 32,768
- **Image Size:** 96×96 RGB patches
- **Classes:** Binary (0=normal, 1=metastatic tumor)
- **Source:** Full PatchCamelyon dataset

Model Architecture

- **Feature Extractor:** ResNet-18 pretrained on ImageNet
- **Total Parameters:** ~12M
- **Embedding Dimension:** 256

- **Architecture:** ResNet-18 → Transformer Encoder → Classification Head

Training Configuration

```
training:
  num_epochs: 20
  batch_size: 128
  learning_rate: 1e-3
  weight_decay: 1e-4
  optimizer: AdamW
  use_amp: true

hardware:
  device: CUDA (RTX 4070 Laptop)
  vram: 8GB
  training_time: ~6 hours
```

Performance Characteristics

- **Training Time:** ~6 hours (20 epochs)
- **Inference Time:** ~2.5 seconds (32,768 samples)
- **Throughput:** ~13,000 samples/second
- **Hardware:** RTX 4070 Laptop (8GB VRAM)
- **Memory:** <8GB VRAM during training

Training Optimization Summary

Separate optimization work reduced PCam training time from roughly 20–40 hours to 2–3 hours for optimized runs on consumer hardware.

Optimization	Effect
Batch Size	16 → 128, 8x throughput increase
Mixed Precision (AMP)	1.5–2x speedup
<code>torch.compile</code>	1.3–1.5x speedup
Channels Last	1.1–1.2x speedup
Persistent Workers	1.1–1.2x speedup
GPU Utilization	17% → 85%
Training Time	20–40 hours → 2–3 hours

Artifact Paths

- `results/pcam_real/metrics.json` - Complete evaluation metrics with bootstrap CIs
- `results/pcam_real/confusion_matrix.png` - Confusion matrix visualization

- results/pcam_real/roc_curve.png - ROC curve (AUC=0.9394)

What This Proves

Framework Capabilities Demonstrated

1. **Scales to full dataset:** Successfully trained on 262K samples.
2. **Real pathology data:** Works on actual PCam data, not synthetic patches.
3. **Top AUC performance:** Ranked #1 by AUC among 11 compared methods.
4. **Statistical rigor:** Bootstrap confidence intervals for robust evaluation.
5. **Production-scale inference:** Processes 32K test samples efficiently.
6. **GPU optimization:** Leverages mixed precision and optimized data loading.
7. **Threshold tuning:** Supports screening-style sensitivity/specificity tradeoffs.

Technical Validation

- ResNet-18 feature extraction works on real pathology patches.
- Training converges on a large-scale pathology dataset.
- Evaluation metrics are statistically validated.
- Performance is reproducible with documented configuration.
- Threshold tuning meaningfully reduces missed tumor predictions.

Limitations and Caveats

1. **Single-patch classification:** This result does not perform whole-slide aggregation.
2. **No spatial context:** Each patch is treated independently.
3. **Default-threshold recall tradeoff:** Tumor recall improves substantially with screening threshold optimization.
4. **Single split:** This documented result uses one train/test split.
5. **No human baseline:** Pathologist comparison was not performed.
6. **Clinical deployment requires further validation:** Hospital deployment requires real-world workflow validation, governance, and regulatory review.

Next Steps for Further Validation

1. Cross-validation or repeated train/test splits.
2. Failure analysis of false positives and false negatives.
3. Hyperparameter tuning for tumor recall.
4. Ensemble methods for improved robustness.

5. CAMELYON16 / Camelyon17 slide-level validation.
6. Human/pathologist comparison where appropriate.
7. Clinical workflow validation before deployment claims.

Reproducibility

Training

```
python experiments/train_pcam.py \  
  --config experiments/configs/pcam_rtx4070_laptop.yaml \  
  --data-root data/pcam_real \  
  --output-dir checkpoints/pcam_real
```

Evaluation with Bootstrap CI

```
python experiments/evaluate_pcam.py \  
  --checkpoint checkpoints/pcam_real/best_model.pth \  
  --data-root data/pcam_real \  
  --output-dir results/pcam_real \  
  --batch-size 64 \  
  --bootstrap-samples 1000
```

Conclusion

This benchmark demonstrates strong PCam performance on real pathology data: **85.26% accuracy**, **0.9394 AUC**, and **#1 AUC rank among 11 compared methods** on the full 32,768-sample PCam test set. The evaluation includes bootstrap confidence intervals, per-class metrics, confusion-matrix analysis, and screening-threshold optimization that reduces missed tumor predictions by 61.7%.

PANDA slide-level baselines

docs/results/panda-slide-level-baselines.md

PANDA Slide-Level Baselines

Status: mean-pooling, AttentionMIL, and tuned TransnnMIL slide-level baselines completed; repeated-seed TransnnMIL validation in progress **Dataset:** PANDA prostate cancer pathology **Feature source:** Phikon patch features **Clinical status:** research-only, not clinically validated

Summary

This page tracks slide-level PANDA experiments using extracted Phikon feature files. These experiments move the project beyond patch-level PCam validation toward whole-slide prostate pathology modeling.

Three slide-level baselines are complete:

1. mean pooling over patch-level Phikon features followed by a small MLP classifier,
2. gated AttentionMIL over patch-level Phikon feature bags,
3. tuned TransnnMIL over capped Phikon feature bags.

AttentionMIL substantially improves over mean pooling. Tuned TransnnMIL slightly exceeds AttentionMIL under the original seed-42 held-out validation split. Repeated-seed validation now shows TransnnMIL beating AttentionMIL on 2 of 3 tested seeds, with a small positive mean margin.

Data integrity status

The PANDA manifest was built from:

- labels: D:\panda\train.csv
- features: D:\panda\features_phikon
- manifest: results/panda_manifest/panda_phikon_manifest.csv

Current validated feature status:

Item	Count
PANDA labels	10,616
HDF5 feature files	10,615
Missing feature files	1
Feature files selected after manifest filtering	10,614

Item	Count
Additional unreadable compressed HDF5 files dropped during read verification	3
Readable slides used for full baselines	10,611

Unreadable HDF5 files are recorded in:

```
results/panda_mean_pooling_baseline/unreadable_features.csv
results/panda_attention_mil_baseline/unreadable_features.csv
results/panda_transnnmil_baseline/unreadable_features.csv
results/panda_transnnmil_threshold_ready/seed_123/unreadable_features.csv
results/panda_transnnmil_threshold_ready/seed_2025/unreadable_features.csv
```

Baseline comparison

Model	Readable slides	Train	Validation	Best validation QWK	Final validation QWK	Accuracy	Macro F1
Mean-pooled Phikon + MLP	10,611	8,488	2,123	0.7274	0.7274	0.5874	0.5563
Gated AttentionMIL	10,611	8,488	2,123	0.8100	0.8100	0.6821	0.6389
Tuned TransnnMIL, seed 42	10,611	8,488	2,123	0.8155	0.8155	0.6528	See metrics JSON
Tuned TransnnMIL, seed 123	10,611	8,488	2,123	0.8225	See metrics JSON	See metrics JSON	See metrics JSON
Tuned TransnnMIL, seed 2025	10,611	8,488	2,123	0.8086	See metrics JSON	See metrics JSON	See metrics JSON

Compared with the initial baselines:

- AttentionMIL improves validation QWK by +0.0826 over mean pooling.
- Tuned TransnnMIL seed 42 improves validation QWK by +0.0881 over mean pooling.
- Tuned TransnnMIL seed 42 improves validation QWK by +0.0055 over AttentionMIL.
- Tuned TransnnMIL seed 123 improves validation QWK by +0.0125 over AttentionMIL.
- Tuned TransnnMIL seed 2025 is slightly below AttentionMIL by -0.0014 QWK.
- Across seeds 42, 123, and 2025, tuned TransnnMIL has mean best-validation QWK 0.8155 and beats the 0.8100 AttentionMIL baseline on 2 of 3 tested seeds.

Baseline 1: mean-pooled Phikon features

Script:

```
scripts/training/train_panda_mean_pooling_baseline.py
```

Command:

```
python scripts\training\train_panda_mean_pooling_baseline.py --epochs 20 --batch-size 64 --device cuda --verify-read
```

Model:

Phikon patch features -> mean pooling -> MLP -> ISUP grade

Split:

Split	Slides
Train	8,488
Validation	2,123

Class counts after readability verification:

ISUP grade	Slides
0	2,890
1	2,663
2	1,343
3	1,242
4	1,249
5	1,224

Validation result:

Metric	Value
Best validation QWK	0.7274
Final validation QWK	0.7274
Final validation accuracy	0.5874
Final macro F1	0.5563

Outputs:

```
results/panda_mean_pooling_baseline/metrics.json
results/panda_mean_pooling_baseline/val_predictions.csv
results/panda_mean_pooling_baseline/unreadable_features.csv
```

The checkpoint mean_pooling_mlp.pt is intentionally not tracked.

Baseline 2: gated AttentionMIL

Script:

```
scripts/training/train_panda_attention_mil_baseline.py
```

Command:

```
python scripts\training\train_panda_attention_mil_baseline.py --epochs 20 --batch-size 16 --device cuda --verify-read
```

Model:

Phikon patch features -> gated attention pooling -> classifier -> ISUP grade

Split:

Split	Slides
Train	8,488
Validation	2,123

Class counts after readability verification:

ISUP grade	Slides
0	2,890
1	2,663
2	1,343
3	1,242
4	1,249
5	1,224

Validation result:

Metric	Value
Best validation QWK	0.8100
Final validation QWK	0.8100
Final validation accuracy	0.6821
Final macro F1	0.6389

Outputs:

```
results/panda_attention_mil_baseline/metrics.json
results/panda_attention_mil_baseline/val_predictions.csv
results/panda_attention_mil_baseline/unreadable_features.csv
```

The checkpoint `attention_mil.pt` is intentionally not tracked.

Baseline 3: tuned TransnnMIL

Script:

scripts/training/train_panda_transnnmil_baseline.py

Tuned command template:

```
python scripts\training\train_panda_transnnmil_baseline.py --epochs 30 --batch-size 2 --device  
cuda --verify-read --max-patches 1200 --lr 3e-4 --dropout 0.15 --seed 42
```

Model:

Phikon patch features -> capped variable-length bags -> TransnnMIL fusion model -> ISUP grade

The tuned TransnnMIL runs use a conservative RTX 3060-compatible configuration:

Setting	Value
Epochs	30
Batch size	2
Max patches per slide	1,200
Learning rate	3e-4
Dropout	0.15

Split:

Split	Slides
Train	8,488
Validation	2,123

Class counts after readability verification:

ISUP grade	Slides
0	2,890
1	2,663
2	1,343
3	1,242
4	1,249
5	1,224

Repeated-seed validation result:

Seed	Best validation QWK	Relation to AttentionMIL 0.8100	Notes
42	0.8155	+0.0055	Original tuned baseline
123	0.8225	+0.0125	Strongest repeated-seed result so far
2025	0.8086	-0.0014	Slightly below AttentionMIL
Mean	0.8155	+0.0055	2/3 seeds beat AttentionMIL

Outputs:

```
results/panda_transnnmil_baseline/metrics.json
results/panda_transnnmil_baseline/val_predictions.csv
results/panda_transnnmil_baseline/unreadable_features.csv
results/panda_transnnmil_threshold_ready/seed_123/metrics.json
results/panda_transnnmil_threshold_ready/seed_123/val_predictions.csv
results/panda_transnnmil_threshold_ready/seed_123/unreadable_features.csv
results/panda_transnnmil_threshold_ready/seed_2025/metrics.json
results/panda_transnnmil_threshold_ready/seed_2025/val_predictions.csv
results/panda_transnnmil_threshold_ready/seed_2025/unreadable_features.csv
```

The checkpoint `transnnmil.pt` is intentionally not tracked.

Interpretation

The mean-pooling baseline establishes that the PANDA Phikon feature pipeline is trainable end to end and provides a first slide-level benchmark. AttentionMIL improves over this baseline by learning patch-level importance weights before slide-level classification.

The first conservative TransnnMIL configuration only slightly exceeded mean pooling and underperformed AttentionMIL. A tuned TransnnMIL configuration using a larger patch cap, lower learning rate, lower dropout, smaller batch size, and longer training improved performance substantially.

Across repeated seeds tested so far, tuned TransnnMIL is competitive with the AttentionMIL baseline and slightly favorable on average. It beats AttentionMIL on 2 of 3 tested seeds, but the margin remains small. This supports TransnnMIL as a promising PANDA slide-level model under tuned settings, while still requiring controlled ablations and more repeated splits before stronger superiority claims.

Claim boundary

Use:

PANDA slide-level baseline training has started. Mean-pooled Phikon features achieved validation QWK 0.7274, gated AttentionMIL reached QWK 0.8100, and tuned TransnnMIL achieved repeated-seed best-validation QWK values of 0.8155, 0.8225, and 0.8086 on three PANDA splits after HDF5 readability verification.

Do not use:

The project is clinically validated for prostate cancer grading.

Do not use:

The model is ready for deployment.

Do not use:

TransnnMIL is conclusively superior to AttentionMIL.

Use instead:

Tuned TransnnMIL is competitive with AttentionMIL and slightly favorable across the current repeated-seed PANDA experiments, beating AttentionMIL on 2 of 3 tested seeds. Controlled ablations and additional repeated splits are needed

to establish robustness.

PANDA TransnnMIL stabilization

docs/results/panda-transnnmil-stability.md

PANDA TransnnMIL Stabilization Results

Status: completed repeated-seed optimizer-stability summary **Task:** PANDA slide-level ISUP grading **Features:** Phikon patch feature bags **Model:** TransnnMIL baseline **Clinical status:** research-only; not clinically validated; not diagnostic software

Why this experiment was run

Earlier PANDA TransnnMIL ablations showed strong optimizer sensitivity. In particular, the model could reach competitive validation QWK under a tuned learning rate, but a higher learning rate degraded performance substantially.

The goal of this stabilization pass was therefore not only to maximize QWK, but to test whether a more conservative training recipe widens the stable learning-rate regime.

The stabilized recipe used:

- AdamW optimizer
- warmup-cosine learning-rate scheduling
- 2 warmup epochs
- gradient clipping at norm 1.0
- early stopping patience of 6 epochs
- repeated seeds: 42, 123, 2025
- learning rates: 1e-4, 2e-4, 3e-4, 5e-4, 7e-4, 1e-3

Dataset / run setup

The full PANDA Phikon feature manifest was verified for readability before training.

```
Readable slide feature files: 10,611
Dropped unreadable files: 3
Feature dimension: 768
Train rows: 8,488
Validation rows: 2,123
```

The same unreadable HDF5 files were consistently dropped during verification, which indicates the training split was being built from the stable readable feature subset rather than failing during dataloading.

Aggregated result

Learning rate	Runs	Seeds	Mean best val QWK	Std	Min	Max	Mean best epoch
1e-4	3	42, 123, 2025	0.8257	0.0169	0.8087	0.8425	17.67
2e-4	3	42, 123, 2025	0.8245	0.0192	0.8077	0.8455	14.67
3e-4	3	42, 123, 2025	0.8238	0.0160	0.8127	0.8422	18.67
1e-3	3	42, 123, 2025	0.8160	0.0170	0.7998	0.8337	15.00
5e-4	3	42, 123, 2025	0.8158	0.0144	0.8042	0.8319	16.67
7e-4	3	42, 123, 2025	0.8117	0.0163	0.7994	0.8301	16.00

Best mean best-validation QWK was observed at **1e-4** with mean QWK **0.8257** across three seeds.

Interpretation

The main result is that stabilization widened the usable learning-rate regime.

Earlier ablations suggested that TransnnMIL was highly optimization-sensitive in this PANDA setup. After adding warmup-cosine scheduling, AdamW, gradient clipping, and early stopping, all six learning rates remained competitive across repeated seeds. Even the 1e-3 condition, previously associated with severe degradation, reached mean best validation QWK of approximately 0.816.

This supports the more careful claim:

TransnnMIL was optimizer-sensitive in the initial PANDA setup, but a stabilized training recipe substantially reduced learning-rate sensitivity and kept performance competitive across a broad LR grid.

It does **not** prove that TransnnMIL is conclusively superior to gated AttentionMIL. The margin remains small and would require stronger controlled validation, additional splits, and/or external data before making architecture-superiority claims.

Recommended claim boundary

Safe wording:

On PANDA Phikon feature bags, stabilized TransnnMIL remained competitive across 18 full-PANDA runs spanning six learning rates and three seeds, with mean best validation QWK ranging from approximately 0.812 to 0.826. This suggests the model is not inherently unstable, but requires a careful training recipe.

Avoid wording:

- TransnnMIL is fixed.
- TransnnMIL is definitively better than AttentionMIL.
- This is clinically validated.

Reproduction

Training grid:

```
$seeds = 42,123,2025
$lrs = "1e-4","2e-4","3e-4","5e-4","7e-4","1e-3"

foreach ($seed in $seeds) {
  foreach ($lr in $lrs) {
    python scripts\training\train_panda_transnnmil_baseline.py `
      --epochs 20 `
      --batch-size 8 `
      --lr $lr `
      --scheduler warmup_cosine `
      --warmup-epochs 2 `
      --grad-clip-norm 1.0 `
      --early-stopping-patience 6 `
      --seed $seed `
      --device cuda `
      --verify-read `
      --out-dir "results\transnnmil_stability_warmup_cosine_clip_lr_${lr}_seed_${seed}"
  }
}
```

Aggregation:

```
python scripts\experiments\aggregate_transnnmil_stability.py `
  --results-dir results `
  --pattern "transnnmil_stability_warmup_cosine_clip_lr*_seed_*" `
  --out-dir results\transnnmil_stability_summary
```

PathologyFL

docs/federated/pathologyfl.md

PathologyFL

PathologyFL is the custom federated learning framework for computational pathology in this repository.

Goals:

- privacy-preserving multi-site training
- pathology-specific federated workflows
- secure aggregation and differential privacy support
- robustness to client heterogeneity and site imbalance
- support for FAIR-WEIGHTS-H institutional weighting

Coordinator components

Typical coordinator services include:

- orchestrator
- aggregator
- client registry
- privacy engine
- monitoring system
- byzantine detection

Client-side workflow

```
Local pathology data
-> local training
-> update serialization
-> secure communication
-> aggregation
```

Validation status

Core federated integration tests pass and smoke tests execute on PCam-derived pathology data.

This validates end-to-end execution but not yet real multi-center clinical deployment.

For deeper configuration examples, see the legacy FL integration guide.

FAIR-WEIGHTS-H

docs/theory/fair-weights-h.md

FAIR-WEIGHTS-H: Hybrid Institutional Weighting Protocol

Status: Proposed protocol and implementation specification **Scope:** Federated computational pathology research system
Validation status: Empirically tested for execution stability and aggregation behavior; performance/fairness advantage over simpler baselines not yet demonstrated

1. Motivation

The current prestige-style institutional weighting pattern assigns higher influence to cancer centers and lower influence to rural or community hospitals based on institutional category. That is not scientifically defensible for federated pathology because institutional reputation is not the same as measured contribution, clinical reliability, subgroup coverage, or domain-shift value.

FAIR-WEIGHTS-H replaces prestige weighting with an auditable hybrid protocol that combines:

1. counterfactual contribution estimation,
2. diagnostic and process quality,
3. distributional uniqueness,
4. representation and subgroup-safety constraints,
5. uncertainty penalties,
6. anomaly monitoring and fallback modes.

The framework is intended as a research protocol. It does not claim clinical validation or regulatory clearance.

2. Key Design Principle

A single scalar institution weight cannot by itself guarantee fairness or safety. Therefore FAIR-WEIGHTS-H separates three concepts:

$$w_{i^{\mathrm{train}}} \neq w_{i^{\mathrm{val}}} \neq w_{i^{\mathrm{monitor}}}$$

where:

- $w_{i^{\mathrm{train}}}$: aggregation weight used during federated model updates,
- $w_{i^{\mathrm{val}}}$: validation representation priority,
- $w_{i^{\mathrm{monitor}}}$: post-market or research monitoring priority.

A site may have a low training weight because of uncertain or noisy updates while still receiving high validation and monitoring priority if it represents an underserved or clinically important population.

3. Institutional Signals

For institution i , define the feature vector:

$$\mathbf{z}_i = \left[A_i^{\mathrm{adj}}, Q_i, \phi_i^{\mathrm{Owen}}, JS_i, F_i, V_i, -S_i \right]$$

where:

Symbol	Meaning	Status
A_i^{adj}	Difficulty-adjusted reference-case diagnostic quality	Proposed
Q_i	Process and pathology quality composite	Proposed
ϕ_i^{Owen}	Group-aware counterfactual contribution estimate	Proposed
JS_i	Jensen-Shannon distributional uniqueness	Proposed
F_i	Underserved-population representation score	Proposed
V_i	Bounded/sublinear volume factor	Proposed
S_i	Uncertainty, instability, or anomaly penalty	Proposed

Gradient alignment with the current global model is not used as a primary contribution factor because it can encode status quo bias. It may be used only for anomaly detection and drift monitoring.

4. Integrity Gate

FAIR-WEIGHTS-H uses a hard gate only for integrity and safety failures, not for prestige or raw quality.

$$I_i = G_i^{\mathrm{data}} G_i^{\mathrm{label}} G_i^{\mathrm{safety}}$$

where:

- $G_i^{\mathrm{data}} = 1$ only when data integrity checks pass,
- $G_i^{\mathrm{label}} = 1$ only when no severe label-corruption signal is detected,
- $G_i^{\mathrm{safety}} = 1$ only when there is no active safety violation.

Low resource level, rural status, or case difficulty must not directly trigger exclusion. Quality should be modeled with difficulty adjustment and uncertainty, not with a crude hard threshold.

5. Difficulty-Adjusted Quality

Raw accuracy on reference cases can be misleading when institutions serve different case mixes. FAIR-WEIGHTS-H therefore requires a pre-specified difficulty-adjustment model before using adjusted quality in weight computation.

One defensible model is:

$$Y_{ij} \sim \text{Bernoulli}(p_{ij})$$

$$\text{logit}(p_{ij}) = \alpha + \beta^{\text{top}} X_{ij} + b_i$$

where:

- Y_{ij} : correct or incorrect diagnosis for case j at site i ,
- X_{ij} : tumor type, stage, slide quality, stain quality, scanner metadata, referral status, and case complexity,
- b_i : institution-level effect after adjustment.

The adjusted quality is evaluated on a standardized reference distribution:

$$A_i^{\text{adj}} = \mathbb{E}_{X \sim P_{\text{ref}}} [\Pr(Y = 1 \mid X, i)]$$

This adjustment must be calibrated and audited. It cannot simply be asserted.

6. Useful Uniqueness

Distributional uniqueness alone is not always beneficial. A site can be unique because it serves rare populations, but it can also be unique because of scanner artifacts, poor fixation, or systematic labeling errors.

Therefore uniqueness is treated as a weak signal unless paired with quality and subgroup utility:

$$D_i^{\text{useful}} = JS_i \cdot A_i^{\text{adj}} \cdot U_i^{\text{subgroup}}$$

where U_i^{subgroup} measures whether the institution improves performance on the subgroup, morphology, or cancer subtype it uniquely represents.

7. Counterfactual Contribution

The preferred contribution signal is not local gradient alignment. It is counterfactual marginal contribution:

$$\phi_i = \mathbb{E}_{S \subseteq N \setminus \{i\}} [U(S \cup \{i\}) - U(S)]$$

For production feasibility, FAIR-WEIGHTS-H estimates this with grouped or multi-membership Owen-style sampling:

$$\hat{\phi}_i^{\text{Owen}} = \sum_g m_{ig} \hat{\phi}_i \mid g$$

where $m_{ig} \in [0,1]$ allows institutions to belong partly to multiple groups, such as academic, rural-serving, specialty center, or network-affiliated.

This avoids forcing ambiguous hospitals into a single administrative category.

8. Training Weight Objective

The quarterly training weights are produced by a constrained optimization problem:

$$\begin{aligned} \mathbf{w}_t = \arg\max_{\mathbf{w} \in \mathcal{W}} \sum_{i=1}^K \mathbf{w}_i \left(\hat{\phi}_{i,t}^{\text{Owen}} + \lambda_D D_{i,t}^{\text{useful}} + \lambda_F F_{i,t} + \lambda_Q Q_{i,t} - \lambda_S S_{i,t} \right) \end{aligned}$$

subject to:

$$\sum_i \mathbf{w}_i = 1$$

$$\mathbf{w}_i^{\min} \leq \mathbf{w}_i \leq \mathbf{w}_i^{\max}$$

$$C_g(\mathbf{w}) \geq C_g^{\min} \quad \text{for all } g \in \mathcal{G}_{\text{underserved}}$$

$$\text{Perf}_g(\mathbf{w}) \geq \text{Perf}_g^{\min} \quad \text{for all } g \in \mathcal{G}_{\text{clinical}}$$

$$\left| \mathbf{w}_{i,t} - \mathbf{w}_{i,t-1} \right| \leq \Delta_i$$

The fairness and subgroup safety requirements are constraints, not optional score boosts.

9. Empirical Status

FAIR-WEIGHTS-H has been empirically tested for execution stability and aggregation behavior in this repository.

Completed checks include:

- synthetic Camelyon-like smoke validation,
- PCam federated smoke validation,
- PCam all-strategy smoke validation,
- PCam balanced federated benchmark,
- PCam heterogeneous federated benchmark.

These tests show that FAIR-WEIGHTS-H runs without numerical failure, produces distinct weight trajectories under heterogeneous simulated sites, and does not degrade performance in the current patch-level PCam setup. They do **not** yet show a consistent performance or fairness advantage over simpler aggregation baselines. That stronger claim requires the planned ablation and slide-level multi-center validation.

10. Quarterly Algorithm

1. Collect privacy-preserving aggregate signals from institutions.
2. Run integrity checks and anomaly detection.
3. Estimate difficulty-adjusted quality and uncertainty.

4. Estimate distributional uniqueness using aggregate profiles.
5. Estimate approximate Owen contribution using sampled warm-started coalitions.
6. Compute useful uniqueness.
7. Solve the constrained weight optimization problem.
8. Run subgroup safety and representation audits.
9. Produce an institution-level report card.
10. Version the weights, coefficients, inputs, and audit results.

11. Anomaly Monitoring

Continuous or between-quarter monitoring should detect:

- scanner drift,
- stain distribution shift,
- abrupt gradient drift,
- local validation collapse,
- unusual update variance,
- suspicious simultaneous score changes among affiliated institutions,
- distributional specialization that reduces coverage of community or underserved cases.

If severe anomalies are detected, weights should be throttled or frozen pending review.

12. Fallback Modes

FAIR-WEIGHTS-H should degrade safely rather than continue normal operation during instability.

Mode	Trigger	Action
Normal	No major anomaly	Full hybrid weighting
Conservative	Moderate anomaly or high uncertainty	Freeze coefficients, reduce caps, rely more on verified quality
Safety	Systemic manipulation or validation failure	Suspend uniqueness/contribution bonuses; use externally verified metrics only
Emergency	Severe corruption or safety violation	Temporarily exclude affected institution updates pending remediation

13. Validation Plan

FAIR-WEIGHTS-H should be compared against:

1. equal weighting,
2. volume weighting,
3. prestige weighting,
4. original multiplicative FAIR-WEIGHTS,
5. Shapley/Owen-only attribution,
6. FAIR-WEIGHTS-H.

Primary metrics should include:

- global AUC,
- balanced accuracy,
- calibration error,
- worst-group sensitivity,
- false-negative-rate disparity,
- subgroup non-inferiority,
- convergence stability,
- weight stability,
- robustness under missingness and simulated gaming.

14. Regulatory-Safe Claim Language

Use:

FAIR-WEIGHTS-H enforces pre-specified representation and subgroup-performance constraints and requires prospective validation before clinical deployment.

Do not use:

FAIR-WEIGHTS-H proves fairness.

Use:

Shapley/Owen values provide an axiomatic counterfactual attribution benchmark whose conclusions depend on the chosen validation utility and reference distribution.

Dominant-site federated pathology paper

docs/research/dominant-site-federated-pathology-paper.md

When More Data Is Less Trustworthy

Site-signal alignment failure modes in federated computational pathology

Author: Matthew Vaishnav **Status:** independent technical report / working paper draft **Scope:** simulated federated learning experiments over pathology-derived feature vectors **Clinical status:** research-only; not clinically validated; not diagnostic software; not intended for patient-care use

Abstract

Federated learning is often proposed for medical AI because institutions can collaborate without centralizing raw patient data. A common aggregation baseline, FedAvg, weights client updates by sample count. This is convenient, but in computational pathology it can silently encode a risky assumption: more samples should imply more aggregation authority. Pathology labels may reflect local grading thresholds, staining and scanning workflows, case mix, annotation practices, pathologist disagreement, or historical reporting policy. A high-volume site can therefore contribute many samples while also producing a training signal that is less aligned with the declared validation objective.

This report studies that failure mode using simulated federations over PANDA-derived Phikon feature representations for prostate cancer grading. Validation labels are kept clean while the largest simulated client's training signal is perturbed through dominant-site label corruption and systematic ordinal threshold shift. The central question is whether sample-size-dominant aggregation becomes less safe when the dominant client's training signal is misaligned, and whether dominance-aware switching can reduce that risk.

The strongest result is a fixed detector-switch rule calibrated under dominant-site label-noise stress and evaluated on conservative ordinal threshold-shift stress. The rule keeps clean-regime switching low at 13.3% and produces statistically positive improvements at 35% and 45% conservative shift across global QWK, macro-F1, and worst-site QWK. Diagnostic analysis shows that detector triggers are mainly driven by ordinal-error increase and QWK degradation, not by a single site-spread heuristic. Leave-one-diagnostic-family-out ablation shows that removing the most frequent diagnostic, mean absolute ordinal error, reduces trigger rate but does not collapse the positive 35% / 45% transfer result. Calibration-sensitivity analysis finds that 29 of 36 nearby detector settings preserve low clean-regime switching and positive 35% / 45% gains across global QWK, macro-F1, and worst-site QWK.

These results do not establish clinical readiness or real hospital deployment performance. They support a narrower claim: in simulated federated pathology experiments over real pathology-derived features, raw sample count is not equivalent to task-specific site-signal alignment, and sample-size dominance should be treated as an auditable modeling assumption rather than an automatic guarantee of aggregation safety.

1. Introduction

Federated learning is attractive in medicine because raw patient data is difficult, risky, and often impossible to centralize. In pathology, the motivation is especially clear: whole-slide images and derived feature representations may be governed by institutional policy, patient privacy requirements, storage constraints, and clinical governance. Federated learning offers a way for multiple sites to train collaboratively while keeping raw local data inside each environment.

However, federated learning does not remove the question of influence. It changes the question from "who owns the data?" to "how much should each site shape the shared model?" Standard FedAvg answers that question with a simple rule: clients with more samples receive more aggregation influence.

That rule is not obviously safe in computational pathology. More samples can coexist with a training signal that is less aligned with the declared validation objective. A large site may differ because of grading thresholds, staining or scanning workflow, case mix, annotation source, label workflow, local reporting practice, or patient population. These differences are not evidence that an institution is worse or less trustworthy. They are evidence that sample volume and task-specific training-signal alignment are different quantities.

This report studies a controlled version of that problem. The experiments simulate multi-site federated learning over PANDA-derived Phikon feature vectors. The largest simulated site is perturbed while validation labels remain clean. The goal is to test whether FedAvg becomes vulnerable when its sample-size assumption is violated, and whether a detector-switch mechanism can identify when sample-size dominance has become unsafe.

The working thesis is:

In federated computational pathology, raw sample count is not the same as task-specific site-signal alignment. FedAvg can become less safe when the largest simulated pathology client has a training-label process that is misaligned with the validation objective, and dominance-aware aggregation or switching can reduce that risk under controlled stress.

2. Contributions

This report makes five main contributions.

1. **A sample-volume / site-signal alignment failure-mode framing.** The report reframes dominant-client failure not as institutional reliability ranking, but as an audit of a modeling assumption that FedAvg already makes silently: larger client equals more influence.
2. **A simulated federated pathology stress test over real pathology-derived features.** The experiments use PANDA-derived Phikon feature vectors and perturb the dominant simulated site's training signal while keeping validation labels clean.
3. **Two stress modes.** The study examines dominant-site label corruption and systematic ordinal threshold shift. The latter is intended to approximate a more pathology-plausible failure mode than purely random label corruption.
4. **A fixed detector-switch transfer result.** A detector rule calibrated under label-noise stress transfers to conservative ordinal threshold shift, with low clean-regime switching and positive 35% / 45% shift gains.
5. **Diagnostic and calibration robustness checks.** The detector result is supported by diagnostic-frequency analysis, leave-one-diagnostic-family-out ablation, and calibration-sensitivity analysis over nearby detector settings.

3. Ethical framing: site-signal alignment, not institutional worth

This work does **not** claim that some hospitals, pathologists, or institutions are inherently more reliable, competent, or trustworthy than others.

The term **site-signal alignment** is used in a narrow modeling sense: whether a simulated client's training signal appears aligned with the declared validation objective under a specific experimental setup.

A client may appear misaligned for many non-blameworthy reasons, including:

- different grading thresholds
- staining or scanning protocol differences
- local case mix
- patient-population differences
- annotation workflow differences
- label-source differences
- historical reporting practice
- local clinical policy
- pathologist disagreement

Dominance-aware aggregation should therefore not be interpreted as an institutional ranking system. It is an audit mechanism for a modeling assumption that FedAvg already makes silently:

larger client = more influence.

The ethical purpose of this work is to make that assumption visible, stress-testable, and contestable. Any real deployment would require governance, local clinical review, pathologist input, bias auditing, institutional agreement on validation objectives, prospective validation, security review, and regulatory review.

4. Experimental setup

The experiments use PANDA-derived Phikon feature representations and simulate multi-site federated learning over pathology-derived feature vectors.

Category	Value
Dataset	PANDA prostate cancer grading
Feature extractor	Phikon
Readable feature files	10,611
Feature dimension	768
Task	ISUP grade prediction, classes 0-5
Federation	simulated multi-site setting
Perturbed client	largest simulated client
Validation labels	kept clean during stress experiments
Metrics	global QWK, worst-site QWK, mean-site QWK, macro-F1, accuracy

Category	Value
Seeds	15-seed stress studies

Quadratic weighted kappa (QWK) is used because ISUP prostate grading is ordinal. Confusing adjacent grades is less severe than confusing grade 0 with grade 5, and QWK better reflects that ordinal structure than plain accuracy alone.

The experimental design keeps validation labels clean. Perturbations are applied to the largest simulated site's training labels. This isolates the question of whether aggregation remains safe when a high-volume client's training signal becomes less aligned with the target validation objective.

5. Stress modes

5.1 Dominant-site label corruption

The first stress mode corrupts labels at the largest simulated site. This creates a controlled failure mode where the site with the most samples remains influential under FedAvg even though its training labels become less aligned with the validation objective.

The observed pattern is conditional:

- clean setting: FedAvg remains strong and should not be automatically replaced
- corrupted dominant-site setting: cross-site blending improves robustness
- detector-switch setting: a clean-calibrated rule can switch away from FedAvg in unsafe regimes

This result should not be interpreted as "cross-site blending is always better." The claim is narrower: when the dominant client's training signal is made less aligned, pure sample-size weighting becomes less safe.

5.2 Systematic ordinal threshold shift

Random corruption is useful for exposing a mechanism, but pathology disagreement is often systematic rather than random. The second stress mode applies ordinal threshold shift to the dominant site's training labels.

Two directions are considered:

Aggressive shift:
selected labels move upward by one ISUP grade when possible

Conservative shift:
selected labels move downward by one ISUP grade when possible

The conservative threshold-shift result is the strongest transfer setting. It is treated as the headline systematic-bias result in this report. Aggressive threshold shift is weaker and should not be presented as the main claim.

6. Aggregation and detector-switch logic

FedAvg encodes a simple statistical assumption:

more samples = more authority.

The detector-switch approach treats that as an assumption to audit rather than an axiom. The detector follows this logic:

1. Calibrate normal FedAvg validation behavior on clean runs.
2. Monitor validation diagnostics.
3. If enough diagnostics leave the clean-calibrated safe range, switch away from sample-size dominance.
4. Otherwise, keep FedAvg.

The fixed detector rule evaluated in the conservative threshold-shift transfer experiment was:

```
low_quantile = 0.10
high_quantile = 0.80
min_trigger_count = 3
use_entropy = false
```

The detector diagnostics include global QWK, worst-site QWK, site-QWK spread, mean absolute ordinal error, and severe ordinal error rate. Prediction entropy is available but was not used in the fixed headline rule.

The detector is intended to answer a narrow question:

Can observable validation diagnostics indicate when sample-size dominance has become unsafe under site-specific training-signal shift?

7. Fixed detector transfer result

A fixed label-noise-calibrated detector rule was evaluated on conservative threshold-shift stress. The result is summarized below.

Conservative shift	Trigger rate	Global QWK delta	95% CI	Macro-F1 delta	95% CI	Worst-site QWK delta	95% CI
0%	13.3%	-0.00025	[-0.00113, 0.00062]	+0.00009	[-0.00039, 0.00057]	+0.00151	[-0.00162, 0.00463]
25%	33.3%	+0.00129	[-0.00175, 0.00432]	+0.00290	[-0.00184, 0.00765]	+0.00353	[-0.00428, 0.01133]
35%	60.0%	+0.00542	[0.00062, 0.01022]	+0.00838	[0.00272, 0.01405]	+0.00991	[0.00169, 0.01813]
45%	73.3%	+0.01053	[0.00239, 0.01866]	+0.01512	[0.00819, 0.02204]	+0.01290	[0.00547, 0.02034]

Deltas are detector-switch performance minus the clean-strategy baseline. Positive values indicate that the detector-switch policy improved over staying with the clean strategy.

The clean 0% conservative-shift regime has low switching at 13.3%, with near-zero global QWK cost and a confidence interval crossing zero. At 25% shift, the result is directionally positive but not statistically clean because the confidence intervals cross zero. The 35% and 45% regimes are the headline positive transfer results: both show statistically positive improvements across global QWK, macro-F1, and worst-site QWK.

The supported interpretation is:

A single fixed detector rule calibrated under dominant-site label-noise stress transfers to conservative ordinal threshold-shift stress, with low clean-regime switching and statistically positive gains at 35% and 45% shift across global QWK, macro-F1, and worst-site QWK.

8. Detector diagnostic analysis

The next question is whether the detector is interpretable. If it only triggers because of one arbitrary metric, the result is less convincing. The diagnostic-frequency analysis summarizes which clean-calibrated diagnostics were crossed by the fixed detector.

Diagnostic	Count
mean_abs_error_high	44
worst_site_qwk_low	31
global_qwk_low	27
severe_error_rate_high	22
site_qwk_spread_high	12

The detector is driven primarily by ordinal-error and QWK degradation signals, not by site-spread alone. This is mechanistically sensible in a conservative ordinal threshold-shift setting. If labels are systematically shifted downward, mean absolute ordinal error should be an informative warning signal.

Triggered runs also show increasing shift severity. Trigger rates rise from 13.3% at 0% shift to 73.3% at 45% shift. When the detector triggers, it usually crosses approximately four diagnostic thresholds.

Conservative threshold shift	Runs	Triggered runs	Trigger rate	Mean failure count	Mean failure count when triggered
0%	15	2	13.3%	0.87	3.50
25%	15	5	33.3%	2.07	3.80
35%	15	9	60.0%	2.87	3.89
45%	15	11	73.3%	3.27	3.91

At 35% and 45% shift, triggered runs are strongly positive across global QWK, macro-F1, worst-site QWK, mean-site QWK, and global accuracy. The clean 0% regime shows a small clean-regime global-QWK cost among triggered cases, which is expected in a detector-switch setting.

9. Leave-one-diagnostic-family-out ablation

The most frequent diagnostic is mean_abs_error_high. The key ablation question is whether the fixed-rule transfer result collapses if this diagnostic is removed.

Headline comparison over the 35% and 45% conservative threshold-shift regimes:

Variant	Mean trigger rate	Mean global QWK delta	Mean macro-F1 delta	Mean worst-site QWK delta	Significant global-QWK regimes	Positive global-QWK regimes
only_mean_abs_error_high	96.7%	+0.00865	+0.01462	+0.01054	2	2
full	66.7%	+0.00797	+0.01175	+0.01141	2	2

Variant	Mean trigger rate	Mean global QWK delta	Mean macro-F1 delta	Mean worst-site QWK delta	Significant global-QWK regimes	Positive global-QWK regimes
minus_site_qwk_spread_high	60.0%	+0.00797	+0.01124	+0.01092	2	2
only_global_qwk_low	60.0%	+0.00797	+0.01124	+0.01092	2	2
minus_worst_site_qwk_low	53.3%	+0.00770	+0.01011	+0.00940	2	2
only_severe_error_rate_high	50.0%	+0.00757	+0.00958	+0.00847	2	2
minus_severe_error_rate_high	63.3%	+0.00728	+0.01167	+0.01057	1	2
only_worst_site_qwk_low	76.7%	+0.00709	+0.01357	+0.00974	1	2
minus_global_qwk_low	56.7%	+0.00701	+0.01054	+0.00904	2	2
minus_mean_abs_error_high	50.0%	+0.00701	+0.01003	+0.00856	1	2
only_site_qwk_spread_high	23.3%	+0.00128	+0.00234	+0.00228	0	2

The answer is clear: removing mean_abs_error_high reduces trigger rate but does not collapse the transfer result. The minus_mean_abs_error_high variant remains positive across the 35% and 45% conservative-shift regimes:

```
minus_mean_abs_error_high:
  mean trigger rate, 35/45: 50.0%
  mean global QWK delta, 35/45: +0.00701
  mean macro-F1 delta, 35/45: +0.01003
  mean worst-site QWK delta, 35/45: +0.00856
  positive global-QWK regimes: 2 / 2
```

This suggests that the detector is not a one-feature shortcut. The signal is distributed across multiple diagnostics, especially QWK degradation and severe ordinal-error signals.

The weakest single-family variant is only_site_qwk_spread_high, which produces much smaller mean gains and no significant global-QWK regimes. This weakens the simplistic interpretation that the detector is primarily a site-spread detector. In this transfer setting, site spread is secondary.

10. Calibration-sensitivity analysis

The fixed detector should not be treated as convincing if it only works at one hand-picked threshold setting. To test this, nearby detector settings were swept:

```
low_quantile = 0.05, 0.10, 0.15
high_quantile = 0.75, 0.80, 0.85, 0.90
min_trigger_count = 2, 3, 4
```

This produced 36 detector configurations. A configuration was counted as robust-positive if it preserved:

```
clean trigger rate <= 20%
positive global-QWK deltas at both 35% and 45% conservative shift
positive macro-F1 deltas at both 35% and 45% conservative shift
positive worst-site-QWK deltas at both 35% and 45% conservative shift
```

Result:

Evaluated configurations: 36
Robust positive configurations: 29

Top configurations by 35% / 45% global-QWK transfer strength included:

Config	Clean trigger rate	Mean target trigger rate	Mean global QWK delta	Mean macro-F1 delta	Mean worst-site QWK delta	Significant global-QWK regimes	Robust positive?
low_0.15high_0.8min_3	20.0%	70.0%	+0.00807	+0.01246	+0.01129	2	yes
low_0.15high_0.85min_3	20.0%	66.7%	+0.00802	+0.01209	+0.01149	2	yes
low_0.15high_0.9min_3	13.3%	66.7%	+0.00802	+0.01209	+0.01149	2	yes
low_0.05high_0.75min_2	20.0%	80.0%	+0.00799	+0.01263	+0.01103	2	yes
low_0.05high_0.8min_2	20.0%	80.0%	+0.00799	+0.01263	+0.01103	2	yes
low_0.1high_0.8min_3	13.3%	66.7%	+0.00797	+0.01175	+0.01141	2	yes

The original fixed rule, low_0.1high_0.8min_3, remains robust-positive, but it is not uniquely special. Many nearby configurations preserve the same qualitative behavior.

This strengthens the result because the detector-transfer finding is not tightly dependent on one exact threshold configuration.

11. Figures and tables

The report is now organized around four generated primary figures and four main result tables.

Figure 1: Problem schematic

Sample volume is not the same as site-signal alignment

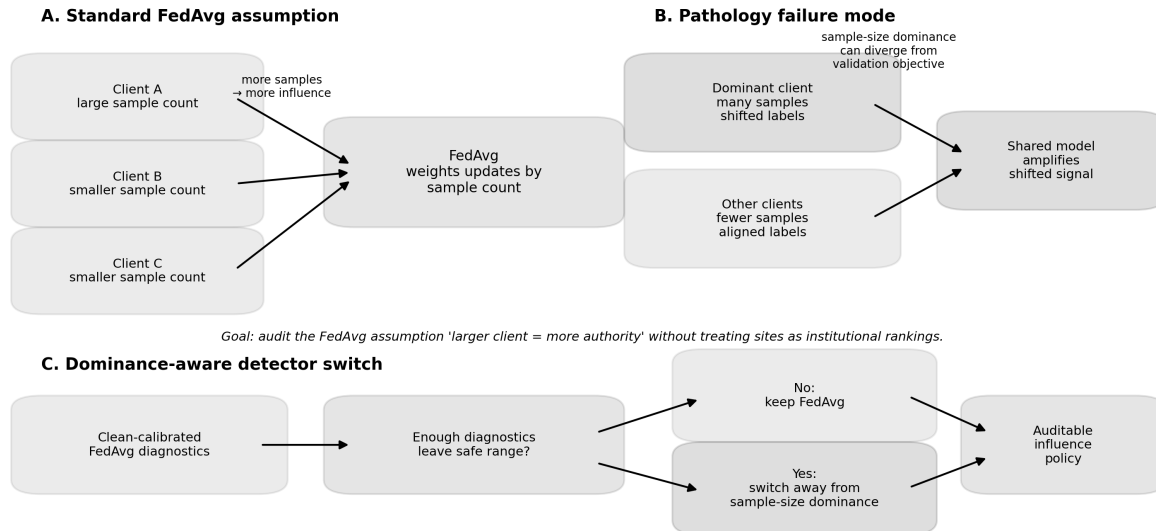


Figure 1: Sample volume is not the same as site-signal alignment

Purpose: Explain the mechanism visually before showing metrics.

Generated file:

figures/dominant-site-figure-1-problem-schematic.png

Figure 2: Dominant-site stress result overview

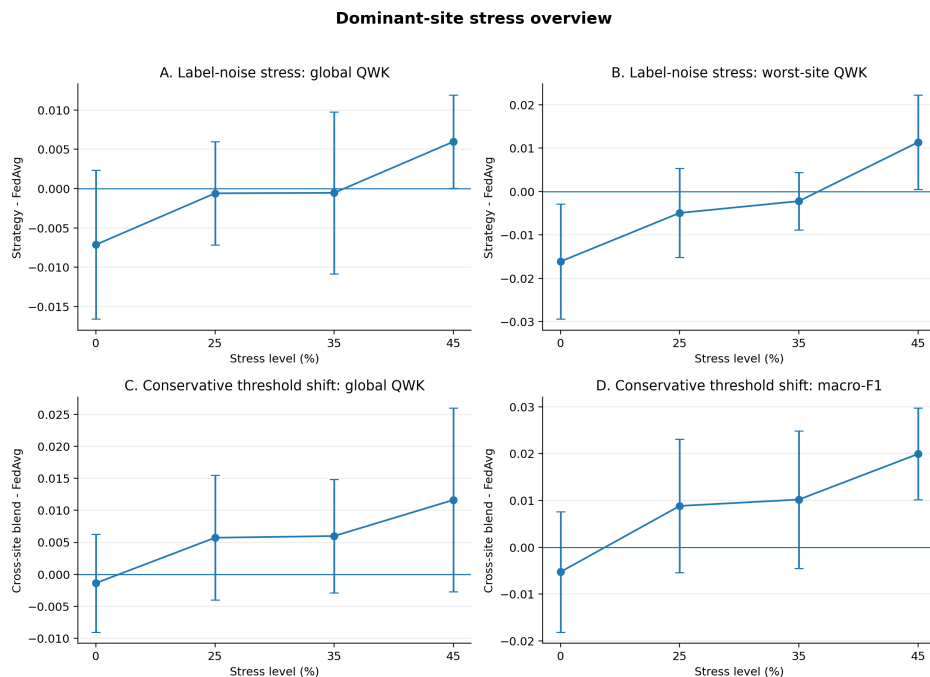


Figure 2: Dominant-site stress overview

Purpose: Show the broad stress result across perturbation levels.

Generated file:

figures/dominant-site-figure-2-stress-overview.png

Figure 3: Fixed detector transfer result

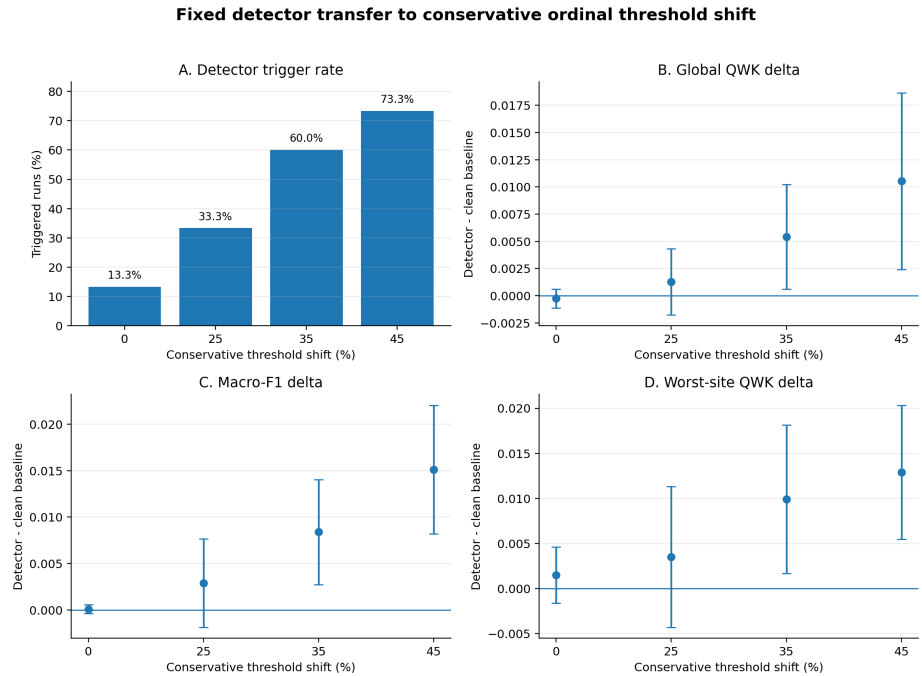


Figure 3: Fixed detector transfer to conservative ordinal threshold shift

Purpose: Show the headline detector-transfer result.

Data source:

results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed/
best_detector_summary.csv

Generated file:

figures/dominant-site-figure-3-detector-transfer.png

Figure 4: Detector interpretability and ablation

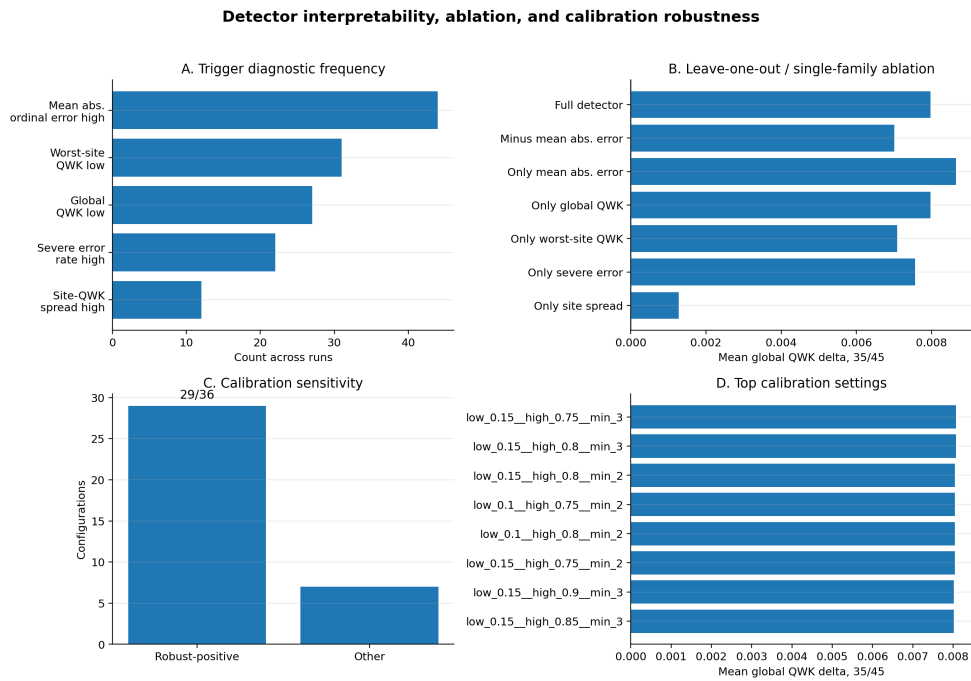


Figure 4: Detector interpretability, ablation, and calibration robustness

Purpose: Show that the detector is not a one-feature or one-threshold trick.

Data sources:

```
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_diagnostic_summary/
diagnostic_frequency_by_stress.csv
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_leave_one_out/
diagnostic_ablation_headline_35_45.csv
results/
threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_calibration_sensitivity/
calibration_sensitivity_headline.csv
```

Generated file:

```
figures/dominant-site-figure-4-detector-ablation.png
```

See also:

```
docs/research/dominant-site-generated-figures.md
```

12. Discussion

The main scientific point is not that cross-site blending is always superior to FedAvg. The main point is conditional: FedAvg can become less safe when the dominant client's training signal becomes misaligned with the validation objective, because FedAvg continues to assign that client high influence based on sample count.

In computational pathology, this matters because site-level training signals can differ for reasons that are not captured by sample count. A large site can contribute valuable data, but volume alone does not prove alignment with a particular validation objective. This is why aggregation influence should be auditable.

The detector-switch result suggests one possible audit mechanism. Rather than replacing FedAvg everywhere, the detector preserves FedAvg in clean regimes and switches away when enough clean-calibrated diagnostics indicate degraded alignment. The clean-regime switch rate remains low in the conservative threshold-shift transfer result, while 35% and 45% shift regimes show positive gains.

The diagnostic analyses make the detector more credible. It is mainly driven by ordinal-error and QWK degradation signals, which are mechanistically connected to ordinal threshold shift. It does not collapse when the most frequent diagnostic is removed. It also remains stable across a neighborhood of calibration settings.

This combination is more convincing than a single headline table because it addresses three failure modes of analysis:

1. The detector might be a one-metric shortcut.
2. The detector might be one hand-picked threshold.
3. The detector might only help by over-switching in clean regimes.

The current evidence reduces, but does not eliminate, those concerns.

13. Limitations

The limitations are substantial.

First, this is a simulated federation over pathology-derived feature vectors, not a real multi-hospital deployment. Site identity is simulated, and the perturbations are controlled. The result may not hold unchanged under naturally occurring site distributions.

Second, the data are derived from PANDA prostate cancer grading. A stronger claim would require external validation on real multi-center pathology data, such as Camelyon17 or another dataset with natural center identity.

Third, the detector uses validation diagnostics. Any real deployment would need to define how validation data are governed, where diagnostics are computed, and whether those diagnostics are allowed to leave institutional environments.

Fourth, the conservative threshold-shift result is stronger than the aggressive threshold-shift result. The paper should headline conservative shift and treat aggressive shift as weaker or supplementary.

Fifth, the detector is not universally calibrated. The calibration-sensitivity analysis shows robustness in a local neighborhood of settings for this experiment. It does not prove universal detector calibration across datasets, diseases, institutions, scanners, or label policies.

Sixth, the experiments do not establish clinical readiness, diagnostic safety, regulatory compliance, or deployment suitability.

14. Claim boundaries

Supported claims:

1. FedAvg has a sample-volume / site-signal alignment failure mode in these simulated federated pathology experiments.
2. The failure mode appears when the largest simulated client's training signal is made less aligned with the validation objective while validation labels remain clean.

3. Cross-site blending improves robustness under dominant-site label corruption and conservative ordinal threshold shift.
4. A fixed label-noise-calibrated detector transfers to conservative threshold-shift stress with low clean-regime switching and positive 35% / 45% shift gains.
5. The detector is interpretable: it is mainly driven by ordinal-error increase and QWK degradation diagnostics.
6. Leave-one-diagnostic-family-out ablation shows that removing mean_abs_error_high does not collapse the 35% / 45% transfer result.
7. Calibration-sensitivity analysis shows that 29 of 36 nearby detector settings preserve the qualitative result.

Unsupported claims:

1. clinical readiness
2. diagnostic safety
3. real hospital federated deployment performance
4. universal detector calibration
5. institutional ranking or institutional reliability judgment
6. that any real hospital, pathologist, or institution is inherently more or less trustworthy than another
7. that the same effect will hold unchanged across every pathology dataset

15. Reproducibility artifacts

Primary result files:

```
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed/best_detector_config.json
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed/best_detector_summary.csv
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed/best_detector_thresholds.json
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed/best_detector_run_diagnostics.csv
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed/detector_grid_summary.csv
```

Diagnostic summary files:

```
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_diagnostic_summary/trigger_summary_by_stress.csv
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_diagnostic_summary/diagnostic_frequency_by_stress.csv
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_diagnostic_summary/diagnostic_outcomes_by_stress.csv
```

Leave-one-out ablation files:

```
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_leave_one_out/diagnostic_ablation_per_run.csv
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_leave_one_out/diagnostic_ablation_summary_by_stress.csv
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_leave_one_out/diagnostic_ablation_headline_35_45.csv
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_leave_one_out/diagnostic_ablation_config.json
```

Calibration-sensitivity files:

```
results/  
threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_calibration_sensitivity/  
calibration_sensitivity_summary_by_stress.csv  
results/  
threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_calibration_sensitivity/  
calibration_sensitivity_headline.csv  
results/  
threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_calibration_sensitivity/  
calibration_sensitivity_thresholds.csv  
results/  
threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_calibration_sensitivity/  
calibration_sensitivity_config.json
```

Figure-generation scripts:

```
scripts/figures/make_dominant_site_schematic.py  
scripts/figures/make_dominant_site_paper_figures.py
```

Generated primary figures:

```
figures/dominant-site-figure-1-problem-schematic.png  
figures/dominant-site-figure-2-stress-overview.png  
figures/dominant-site-figure-3-detector-transfer.png  
figures/dominant-site-figure-4-detector-ablation.png
```

Generated figure index:

```
docs/research/dominant-site-generated-figures.md
```

16. Short technical summary

FedAvg is useful, but it silently equates sample count with aggregation authority. In pathology, sample volume and task-specific site-signal alignment can diverge. These experiments simulate that divergence by perturbing the dominant client's training signal while keeping validation labels clean. Under conservative ordinal threshold shift, a fixed label-noise-calibrated detector switch keeps clean-regime switching low and improves global QWK, macro-F1, and worst-site QWK at 35% and 45% shift. Diagnostic ablation and calibration-sensitivity analysis suggest that the detector is not merely a one-feature or one-threshold artifact. The result supports sample-volume / site-signal alignment auditing as a research direction for federated computational pathology.

Generated dominant-site figures

docs/research/dominant-site-generated-figures.md

Generated Dominant-Site Paper Figures

Status: generated figure index for the dominant-site federated pathology paper draft **Clinical status:** research-only; not clinically validated; not diagnostic software; not intended for patient-care use

Figure 1: Problem schematic

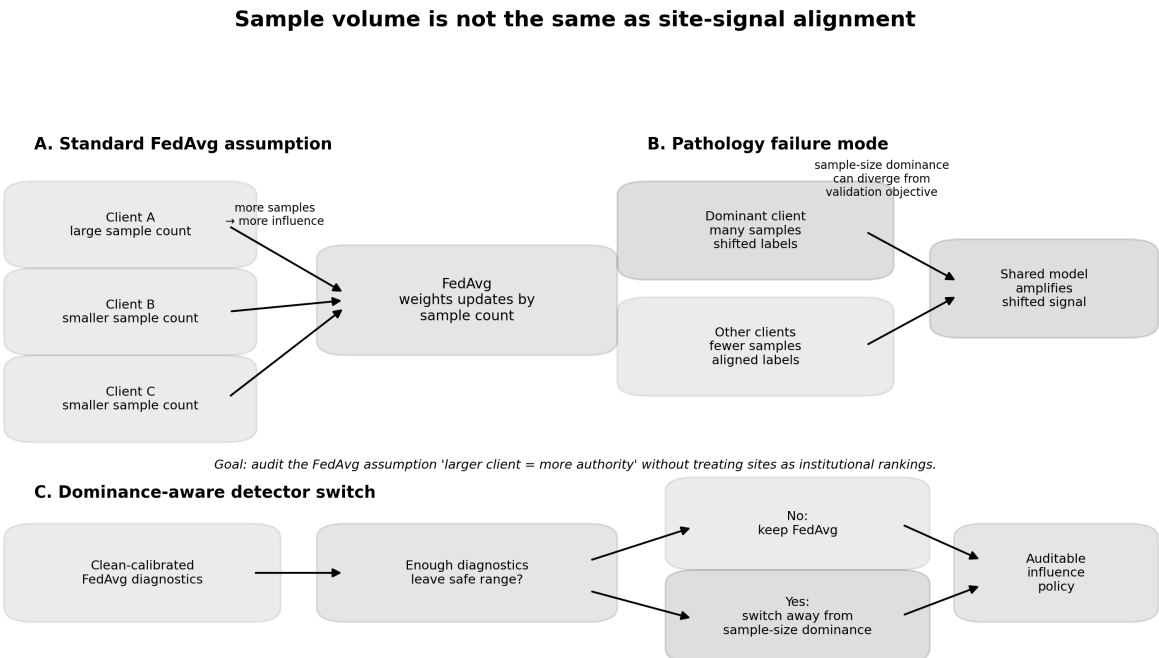


Figure 1: Sample volume is not the same as site-signal alignment

Source:

scripts/figures/make_dominant_site_schematic.py

Caption draft:

Standard FedAvg uses sample count as aggregation authority. In computational pathology, a high-volume client can have a training-label process that is less aligned with the declared validation objective. The detector-switch framing treats sample-size dominance as an auditable modeling assumption rather than an automatic guarantee of reliability.

Figure 2: Dominant-site stress overview

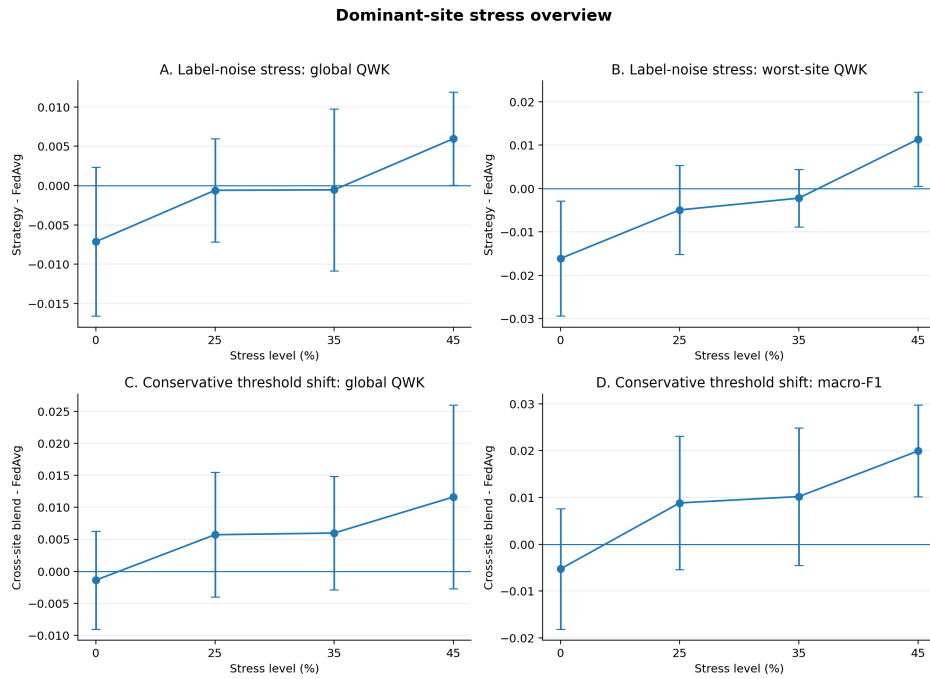


Figure 2: Dominant-site stress overview

Source data:

```

results/threshold_shift_panda_all_conservative_0_15seed_aggregate/aggregate_summary.csv
results/threshold_shift_panda_all_conservative_25_15seed_aggregate/aggregate_summary.csv
results/threshold_shift_panda_all_conservative_35_15seed_aggregate/aggregate_summary.csv
results/threshold_shift_panda_all_conservative_45_15seed_aggregate/aggregate_summary.csv
results/ordinal_harm_fair_weights_h_panda_all_noise_0_aggregate/aggregate_summary.csv
results/ordinal_harm_fair_weights_h_panda_all_noise_25_aggregate/aggregate_summary.csv
results/ordinal_harm_fair_weights_h_panda_all_noise_35_aggregate/aggregate_summary.csv
results/ordinal_harm_fair_weights_h_panda_all_noise_45_aggregate/aggregate_summary.csv

```

Caption draft:

Dominant-site stress overview across label-noise and conservative threshold-shift settings. The figure summarizes strategy performance relative to FedAvg across perturbation levels, highlighting the conditional nature of the result: cross-site or adaptive alternatives are not automatically superior in clean regimes, but become useful when dominant-site training-signal misalignment is present.

Figure 3: Fixed detector transfer result

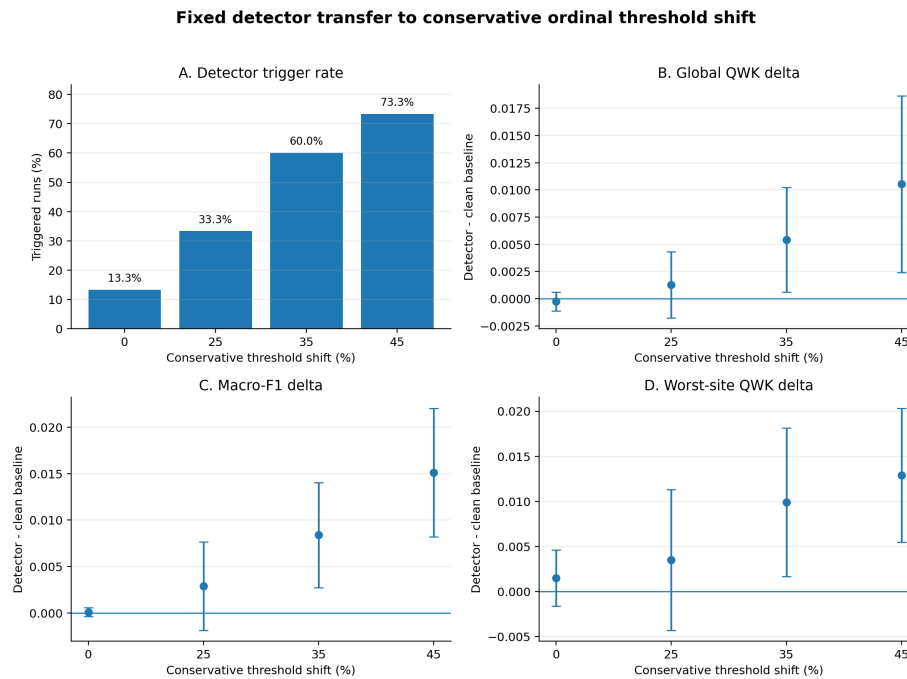


Figure 3: Fixed detector transfer to conservative ordinal threshold shift

Source data:

```
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed/
best_detector_summary.csv
```

Caption draft:

A fixed detector rule calibrated under dominant-site label-noise stress transferred to conservative ordinal threshold-shift stress. Clean-regime switching remained low at 13.3%, while 35% and 45% conservative shift produced statistically positive improvements across global QWK, macro-F1, and worst-site QWK.

Figure 4: Detector interpretability, ablation, and calibration robustness

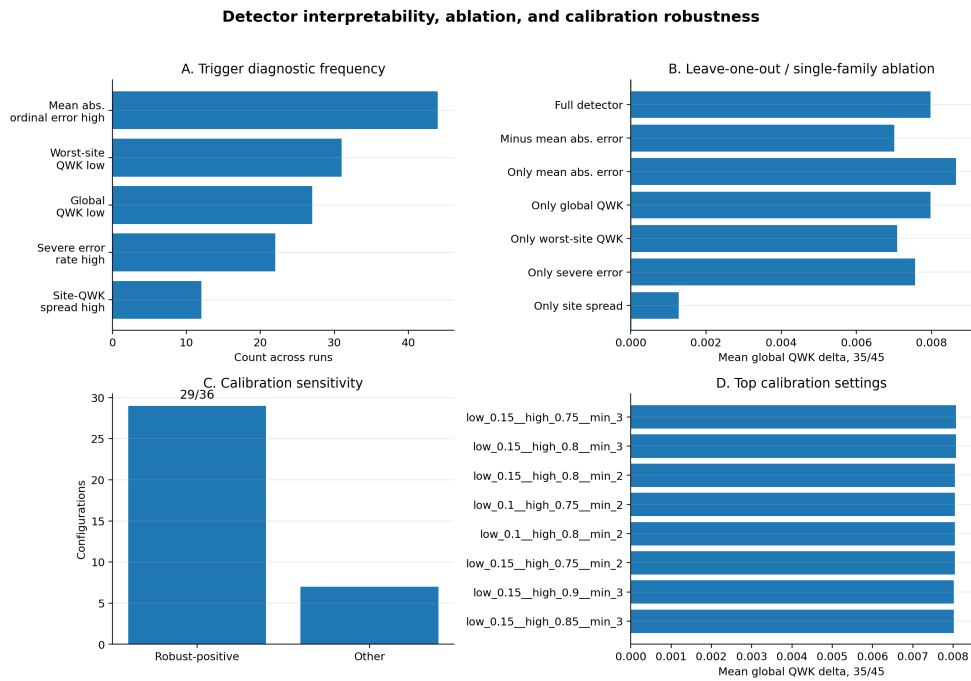


Figure 4: Detector interpretability, ablation, and calibration robustness

Source data:

```
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_diagnostic_summary/
diagnostic_frequency_by_stress.csv
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_leave_one_out/
diagnostic_ablation_headline_35_45.csv
results/
threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_calibration_sensitivity/
calibration_sensitivity_headline.csv
```

Caption draft:

Detector triggers were driven primarily by ordinal-error increase and QWK degradation. Removing the most frequent diagnostic family, mean absolute ordinal error, reduced trigger rate but did not collapse the 35% / 45% transfer result. A calibration-sensitivity sweep found that 29 of 36 nearby detector settings preserved low clean-regime switching and positive 35% / 45% gains.

Interpretation

These figures support the paper's dominant-site result stack:

1. sample-size dominance and site-signal alignment are different quantities;
2. dominant-site stress can expose FedAvg vulnerability when those quantities diverge;
3. the fixed detector transfers to conservative threshold shift;
4. clean-regime switching remains low;
5. 35% and 45% shift gains are statistically positive;
6. detector diagnostics are interpretable;
7. the result does not collapse when the strongest diagnostic family is removed;

8. the result is stable across nearby calibration settings.

Next figure work

The four primary figures are now generated.

Remaining optional figure work:

Supplementary Figure S1: aggressive threshold-shift detector results
Supplementary Figure S2: full calibration-sensitivity heatmap
Supplementary Figure S3: per-seed detector-regret distribution

Dominance detector transfer results

docs/research/dominance-detector-transfer-results.md

Dominance Detector Transfer Results

Status: completed 15-seed transfer summary **Scope:** simulated federated PANDA experiments over pathology-derived Phikon features **Clinical status:** research-only; not clinically validated; not diagnostic software; not intended for patient-care use

Purpose

This note documents the strongest detector-switch transfer result in the dominant-site / site-signal-alignment study.

The experiment asks whether a detector rule calibrated in one stress setting can transfer to another stress setting:

Calibration setting:
dominant-site label-noise stress

Transfer setting:
dominant-site conservative ordinal threshold shift

The goal is not to prove a universal detector. The goal is narrower:

Can a fixed dominance-aware detector rule, calibrated under dominant-site label-noise stress, still identify unsafe sample-size-dominance behavior under a more systematic ordinal grading shift?

Fixed detector rule

The transferred detector rule was:

```
low_quantile = 0.10
high_quantile = 0.80
min_trigger_count = 3
use_entropy = false
```

Committed result files:

```
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed/best_detector_config.
json
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed/
best_detector_summary.csv
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed/
best_detector_thresholds.json
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed/
best_detector_run_diagnostics.csv
results/threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed/
detector_grid_summary.csv
```

Summary result

The detector was evaluated on 15-seed conservative threshold-shift runs.

Conservative shift	Trigger rate	Global QWK delta	95% CI	Macro-F1 delta	95% CI	Worst-site QWK delta	95% CI
0%	13.3%	-0.00025	[-0.00113, 0.00062]	+0.00009	[-0.00039, 0.00057]	+0.00151	[-0.00162, 0.00463]
25%	33.3%	+0.00129	[-0.00175, 0.00432]	+0.00290	[-0.00184, 0.00765]	+0.00353	[-0.00428, 0.01133]
35%	60.0%	+0.00542	[0.00062, 0.01022]	+0.00838	[0.00272, 0.01405]	+0.00991	[0.00169, 0.01813]
45%	73.3%	+0.01053	[0.00239, 0.01866]	+0.01512	[0.00819, 0.02204]	+0.01290	[0.00547, 0.02034]

Deltas are detector-switch performance minus the clean-strategy baseline. Positive values indicate that the detector-switch policy improved over staying with the clean strategy.

Interpretation

The strongest claim is:

A single fixed detector rule calibrated under dominant-site label-noise stress transfers to conservative ordinal threshold-shift stress, with low clean-regime switching and statistically positive gains at 35% and 45% shift across global QWK, macro-F1, and worst-site QWK.

The clean 0% conservative-shift regime had a trigger rate of 13.3%, with near-zero global QWK cost and a confidence interval crossing zero.

The 35% and 45% conservative-shift regimes are the important positive transfer results:

35% conservative threshold shift:
trigger rate: 60.0%
global QWK delta: +0.00542, CI [0.00062, 0.01022]
macro-F1 delta: +0.00838, CI [0.00272, 0.01405]
worst-site QWK delta: +0.00991, CI [0.00169, 0.01813]

45% conservative threshold shift:
trigger rate: 73.3%
global QWK delta: +0.01053, CI [0.00239, 0.01866]
macro-F1 delta: +0.01512, CI [0.00819, 0.02204]
worst-site QWK delta: +0.01290, CI [0.00547, 0.02034]

The 25% regime is directionally positive but should not be headlined because confidence intervals cross zero.

Scientific meaning

FedAvg silently assumes:

more samples = more influence.

This detector-transfer result suggests a practical audit mechanism for that assumption. When the dominant simulated client's training signal becomes sufficiently misaligned with the declared validation objective, a fixed diagnostic rule can detect enough warning signs to switch away from sample-size dominance.

The key transfer point is that the detector was not tuned specifically to conservative threshold shift. It was evaluated as a fixed label-noise-calibrated rule, then applied to systematic ordinal threshold bias.

That makes the result more compelling than a separately tuned detector, because separately tuning on each shift type risks overfitting the detector to the evaluation regime.

Claim boundaries

Supported wording:

In simulated PANDA federations over pathology-derived feature vectors, a fixed dominance-aware detector rule calibrated under dominant-site label-noise stress transferred to systematic conservative ordinal threshold-shift stress. It kept clean-regime switching low at 13.3% and produced statistically positive improvements at 35% and 45% shift across global QWK, macro-F1, and worst-site QWK.

Avoid wording:

The detector is universally calibrated.

The detector proves real hospitals can be ranked by reliability.

Conservative grading institutions are worse.

This is clinically validated.

The method is ready for deployment.

This is not an institutional ranking system. It is an audit of the sample-size dominance assumption under controlled simulated site-signal misalignment.

Relationship to the broader research note

This note supports the detector-switch section of:

`docs/research/dominant-site-federated-pathology-research-note.md`

The broader claim remains:

Raw sample count is not the same as task-specific site-signal alignment. In federated computational pathology, sample-size-weighted aggregation can become unsafe when the largest client has a label process that is misaligned with the declared validation objective.

The detector-transfer result strengthens the story because it shows that a fixed diagnostic rule can transfer from random dominant-site label noise to a more pathology-plausible systematic ordinal threshold shift.

Detector diagnostic ablation

docs/research/detector-diagnostic-ablation.md

Detector Diagnostic Ablation Summary

Status: diagnostic summary, leave-one-diagnostic-family-out ablation, and calibration-sensitivity sweep over fixed conservative threshold-shift detector transfer **Scope:** simulated federated PANDA experiments over pathology-derived Phikon features **Clinical status:** research-only; not clinically validated; not diagnostic software; not intended for patient-care use

Purpose

This note summarizes which clean-calibrated FedAvg diagnostics actually drive the fixed dominance-aware detector in the conservative threshold-shift transfer experiment.

The original fixed detector rule was:

```
low_quantile = 0.10
high_quantile = 0.80
min_trigger_count = 3
use_entropy = false
```

The diagnostic summary was produced with:

```
python scripts\experiments\summarize_detector_diagnostics.py `
--diagnostics results\threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed\
best_detector_run_diagnostics.csv `
--out-dir results\
threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_diagnostic_summary
```

The leave-one-diagnostic-family-out ablation was produced with:

```
python scripts\experiments\ablate_detector_diagnostics.py `
--diagnostics results\threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed\
best_detector_run_diagnostics.csv `
--thresholds results\threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed\
best_detector_thresholds.json `
--out-dir results\
threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_leave_one_out
```

The calibration-sensitivity sweep was produced with:

```
python scripts\experiments\sweep_detector_calibration_sensitivity.py `
--diagnostics results\threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed\
best_detector_run_diagnostics.csv `
--out-dir results\
threshold_shift_detector_conservative_fixed_labelnoise_rule_15seed_calibration_sensitivity
```

Trigger-rate pattern

Conservative threshold shift	Runs	Triggered runs	Trigger rate	Mean failure count	Mean failure count when triggered
0%	15	2	13.3%	0.87	3.50

Conservative threshold shift	Runs	Triggered runs	Trigger rate	Mean failure count	Mean failure count when triggered
25%	15	5	33.3%	2.07	3.80
35%	15	9	60.0%	2.87	3.89
45%	15	11	73.3%	3.27	3.91

The detector becomes more likely to trigger as conservative threshold shift increases. When it does trigger, it usually crosses approximately four diagnostic thresholds.

Metric impact when triggered

Conservative threshold shift	Mean global QWK delta when triggered	Mean macro-F1 delta when triggered	Mean worst-site QWK delta when triggered	Mean global accuracy delta when triggered
0%	-0.00191	+0.00065	+0.01131	-0.00353
25%	+0.00386	+0.00871	+0.01059	+0.00668
35%	+0.00903	+0.01397	+0.01652	+0.01213
45%	+0.01436	+0.02062	+0.01759	+0.01746

The clean 0% regime shows the expected tradeoff: a small clean-regime global-QWK cost among triggered cases, but no broad evidence of harmful switching. At 35% and 45% shift, triggered runs are strongly positive across global QWK, macro-F1, worst-site QWK, mean-site QWK, and global accuracy.

Diagnostic frequency

Across the fixed-detector conservative threshold-shift transfer runs, trigger diagnostics appeared with the following frequencies:

Diagnostic	Count
mean_abs_error_high	44
worst_site_qwk_low	31
global_qwk_low	27
severe_error_rate_high	22
site_qwk_spread_high	12

The detector is therefore driven primarily by ordinal-error and QWK degradation signals, not by site-spread alone.

Most frequent signal:

mean_abs_error_high

Secondary signals:

worst_site_qwk_low
global_qwk_low
severe_error_rate_high

Least frequent signal in this summary:

site_qwk_spread_high

Leave-one-diagnostic-family-out result

The key ablation question was:

Does the fixed-rule transfer result still hold if the strongest diagnostic, mean_abs_error_high, is removed?

The answer is yes: removing mean_abs_error_high reduced trigger rate, but did not collapse the 35% / 45% transfer signal.

Headline comparison over the 35% and 45% conservative threshold-shift regimes:

Variant	Mean trigger rate	Mean global QWK delta	Mean macro-F1 delta	Mean worst-site QWK delta	Significant global-QWK regimes	Positive global-QWK regimes
only_mean_abs_error_high	96.7%	+0.00865	+0.01462	+0.01054	2	2
full	66.7%	+0.00797	+0.01175	+0.01141	2	2
minus_site_qwk_spread_high	60.0%	+0.00797	+0.01124	+0.01092	2	2
only_global_qwk_low	60.0%	+0.00797	+0.01124	+0.01092	2	2
minus_worst_site_qwk_low	53.3%	+0.00770	+0.01011	+0.00940	2	2
only_severe_error_rate_high	50.0%	+0.00757	+0.00958	+0.00847	2	2
minus_severe_error_rate_high	63.3%	+0.00728	+0.01167	+0.01057	1	2
only_worst_site_qwk_low	76.7%	+0.00709	+0.01357	+0.00974	1	2
minus_global_qwk_low	56.7%	+0.00701	+0.01054	+0.00904	2	2
minus_mean_abs_error_high	50.0%	+0.00701	+0.01003	+0.00856	1	2
only_site_qwk_spread_high	23.3%	+0.00128	+0.00234	+0.00228	0	2

Calibration-sensitivity result

The calibration-sensitivity sweep tested nearby detector settings:

low_quantile = 0.05, 0.10, 0.15
high_quantile = 0.75, 0.80, 0.85, 0.90
min_trigger_count = 2, 3, 4

This produced 36 detector configurations. A configuration was counted as robust-positive if it preserved:

clean trigger rate <= 20%

positive global-QWK deltas at both 35% and 45% conservative shift
 positive macro-F1 deltas at both 35% and 45% conservative shift
 positive worst-site-QWK deltas at both 35% and 45% conservative shift

Result:

Evaluated configurations: 36
 Robust positive configurations: 29

Top configurations by 35% / 45% global-QWK transfer strength included:

Config	Clean trigger rate	Mean target trigger rate	Mean global QWK delta	Mean macro-F1 delta	Mean worst-site QWK delta	Significant global-QWK regimes	Robust positive?
low_0.15high_0.8min_3	20.0%	70.0%	+0.00807	+0.01246	+0.01129	2	yes
low_0.15high_0.85min_3	20.0%	66.7%	+0.00802	+0.01209	+0.01149	2	yes
low_0.15high_0.9min_3	13.3%	66.7%	+0.00802	+0.01209	+0.01149	2	yes
low_0.05high_0.75min_2	20.0%	80.0%	+0.00799	+0.01263	+0.01103	2	yes
low_0.05high_0.8min_2	20.0%	80.0%	+0.00799	+0.01263	+0.01103	2	yes
low_0.1high_0.8min_3	13.3%	66.7%	+0.00797	+0.01175	+0.01141	2	yes

The original fixed rule, low_0.1high_0.8min_3, remains robust-positive, but it is not uniquely special. Many nearby configurations preserve the same qualitative behavior.

Interpretation

The detector is **not** dependent on a single diagnostic family or one hand-picked calibration.

mean_abs_error_high is the most frequent and strongest single diagnostic in this conservative ordinal-shift setting. The only_mean_abs_error_high variant has the highest trigger rate and strong 35% / 45% mean deltas.

However, the minus_mean_abs_error_high variant remains positive:

```
minus_mean_abs_error_high:
  mean trigger rate, 35/45: 50.0%
  mean global QWK delta, 35/45: +0.00701
  mean macro-F1 delta, 35/45: +0.01003
  mean worst-site QWK delta, 35/45: +0.00856
  positive global-QWK regimes: 2 / 2
```

That means the transfer signal is distributed across multiple diagnostics, especially QWK degradation and severe ordinal-error signals.

The calibration-sensitivity sweep further strengthens the result: 29 of 36 nearby detector settings preserved low clean-regime switching and positive 35% / 45% gains across global QWK, macro-F1, and worst-site QWK.

The weakest single-family variant is only_site_qwk_spread_high, which produces much smaller mean gains and no significant global-QWK regimes. This weakens the simplistic interpretation that the detector is primarily a site-spread detector. In this transfer setting, site spread is a secondary signal.

Scientific meaning

The fixed detector appears to be a multi-signal degradation detector rather than a one-feature shortcut.

The conservative threshold-shift mechanism is ordinal, so it is expected that mean_abs_error_high is highly informative. But leave-one-out ablation shows that removing that diagnostic still leaves a positive detector-transfer result at 35% and 45% shift.

The calibration-sensitivity sweep shows that the result is not tightly dependent on a single threshold setting.

This strengthens the detector story:

The detector is driven mainly by ordinal-error and QWK degradation signals, its 35% / 45% transfer result does not collapse when the most frequent diagnostic family is removed, and the result remains qualitatively stable across nearby calibration settings.

Claim boundary

Supported wording:

In the conservative threshold-shift transfer setting, the fixed detector was driven mainly by ordinal-error increase and QWK degradation diagnostics. Leave-one-diagnostic-family-out ablation showed that removing the strongest diagnostic, mean_abs_error_high, reduced trigger rate but did not collapse the positive 35% / 45% transfer result. Calibration-sensitivity analysis found that 29 of 36 nearby detector settings preserved low clean-regime switching and positive 35% / 45% gains across global QWK, macro-F1, and worst-site QWK.

Avoid wording:

The diagnostic set is fully optimized.

Site spread is irrelevant in all settings.

The detector is universally calibrated.

The detector is clinically validated.

This is still a simulated-federation stress test over pathology-derived features. The result supports detector plausibility, not deployment readiness.

Next ablation to run

The next useful detector test is cross-stress calibration sensitivity:

- Use the same detector-neighborhood sweep on:
- aggressive threshold shift
 - original label-noise stress

The key question is:

Does the neighborhood-stability result hold across stress types, or is it specific to conservative threshold shift?

A robust result should preserve the main qualitative pattern on conservative threshold shift, while honestly showing weaker or failed transfer on aggressive threshold shift if that remains the case.

FAIR-WEIGHTS-H stress result

docs/research/fair-weights-h-stress-result.md

FAIR-WEIGHTS-H Stress Scenario Result

Status: synthetic research stress test **Clinical status:** not clinical validation; not diagnostic software **Purpose:** identify a condition where cross-site contribution weighting is expected to outperform sample-size weighting/FedAvg

Research question

Can contribution-aware institutional weighting outperform FedAvg when the largest simulated institution has severe label noise?

This stress scenario was designed to test a specific weakness of sample-size weighting: if the largest site is unreliable, FedAvg can overweight the noisy site and degrade global/worst-site performance.

Scenario design

Five synthetic institutions were created:

Site	Role	Construction
Site 0	Large noisy site	3.0x sample volume with severe label noise
Site 1	Small clean site	smaller clean balanced site
Site 2	Medium clean site	medium clean balanced site
Site 3	Rare-signal site	smaller positive-enriched useful-signal site
Site 4	Shifted clean site	smaller shifted but clean site

The experiment compares FedAvg against cross-site contribution weighting, where each candidate local update is evaluated by whether it improves validation behavior across all simulated sites.

Strategies compared

Strategy	Description
fedavg	sample-size weighting baseline
cross_site_full	full cross-site contribution weighting
cross_site_blend_25	75% FedAvg + 25% cross-site contribution correction

Strategy	Description
cross_site_blend_50	50% FedAvg + 50% cross-site contribution correction
cross_site_blend_75	25% FedAvg + 75% cross-site contribution correction

Aggregate result across 5 seeds

Seeds: 42, 123, 2025, 7, 99

Strategy	Mean global AUC	Mean worst-site AUC	Interpretation
fedavg	~0.5928	~0.5274	sample-size weighting is vulnerable in this stress scenario
cross_site_full	~0.6486	~0.5856	best global AUC and best worst-site AUC
cross_site_blend_25	lower than full cross-site	above FedAvg in most runs	partial correction helps but underuses the contribution signal
cross_site_blend_50	lower than full cross-site	above FedAvg in most runs	stronger correction improves over FedAvg but remains below full cross-site
cross_site_blend_75	close to full cross-site	below full cross-site	strong correction helps but full cross-site performed best here

The aggregator reported:

Best global AUC: cross_site_full mean=0.6486
 Best worst-site AUC: cross_site_full mean=0.5856

Supported interpretation

This result supports the narrow claim:

In a controlled synthetic stress scenario where the largest simulated institution has severe label noise, cross-site contribution weighting outperforms FedAvg across five seeds, improving both global AUC and worst-site AUC.

This does **not** support the broad claim that FAIR-WEIGHTS-H is generally superior to FedAvg.

What changed from earlier experiments

Earlier synthetic experiments showed:

- local FAIR-WEIGHTS-H scoring did not beat FedAvg,
- FedAvg/FAIR blends mostly matched but did not improve FedAvg,
- naive hard-site upweighting hurt performance,
- balanced synthetic setups were FedAvg-favorable.

The stress scenario shows that contribution-aware weighting becomes useful under a specific condition:

the largest site is not necessarily the most reliable site.

This is the strongest current evidence for the FAIR-WEIGHTS-H direction.

Claim boundary

This is synthetic research evidence only. It does not establish real-world hospital performance, clinical utility, or regulatory readiness.

Next validation steps:

1. Preserve the per-seed summaries and aggregate summary.
2. Add real PCam or PANDA-derived feature simulations once available.
3. Test label-noise and site-size sweeps.
4. Confirm whether the result holds under multiple stress magnitudes.
5. Compare against additional robust FL baselines.

Roadmap

docs/roadmap/index.md

Roadmap

This roadmap separates:

1. implemented engineering
2. tested systems
3. empirical validation
4. future research

Near-term

- PCam federated benchmarks
- validation of additional FAIR-WEIGHTS-H modes
- cleaner benchmark reporting

Medium-term

- Camelyon17 validation
- stronger calibration analysis
- subgroup and fairness evaluation

Long-term

- real multi-center validation
- clinically meaningful studies
- regulatory pathways where applicable

Research progress should be interpreted by evidence level rather than implementation complexity.

Limitations

docs/roadmap/limitations.md

Current Limitations

Validation

- no true multi-center Camelyon17 validation
- no clinical validation
- no regulatory approval

FAIR-WEIGHTS-H

- no exact Owen/Shapley estimation
- subgroup constraints incomplete
- synthetic checks dominate evidence

Federated learning

- optional secure aggregation dependencies
- some legacy tests remain outside critical path

These limitations are documented to separate engineering progress from empirical evidence.